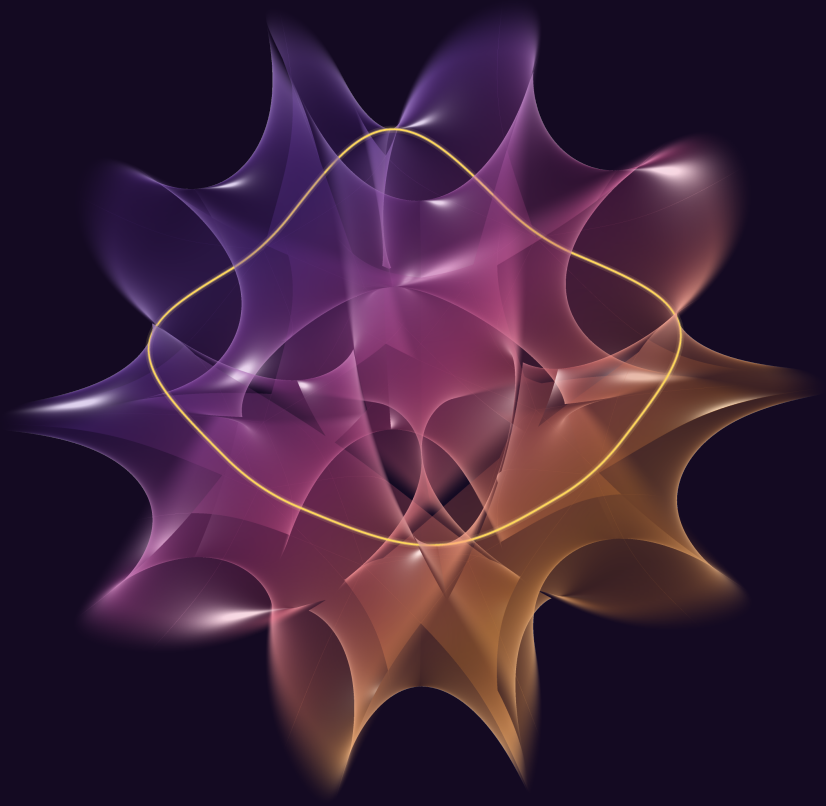


Introduction to String Theory

Part II: String compactifications

Lecture notes on Module 1

University of Bologna, academic year 2026/27

Editor's note

These are notes for Module 1 of the "Introduction to String Theory" course at the University of Bologna. Apart from this page, they were written entirely by Claude Opus 5.5 from scans of handwritten lecture notes taken by me and other students in 2026. They are an independent reconstruction, not a record of the lectures. The lecturer has neither seen nor endorsed them, and they may depart from the lectures in content, emphasis and conventions. Any errors are mine (and Claude's).

I hope these notes prove useful, although I doubt they will hold much value in the years to come, when far more powerful models will be available.

This leads me to some (somewhat uncalled-for and unprofessional) reflections on the strange times I am living through. I remember the moment, a few years ago, when a new OpenAI model turned out to be a better coder than me. Given how fast these models were improving, I knew I would never catch up. Now I feel the same way about physics. I find it hard to believe that I will ever solve a physics problem that an AI could not solve faster and more cheaply. This has made me genuinely sad for months now.

But, to complicate matters, mixed in with that sadness is real excitement. What will these systems teach us about the deepest mysteries in physics? About quantum gravity and string theory? Which problems will they solve? Perhaps the most important practical question a young graduate student can now ask is: "How can I position myself to understand the output of the equivalent of a thousand Wittens, working nonstop at fifty times human speed?" I hope these lecture notes can help with that task.

LazyLagrangian
6 November 2026

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Notation and conventions

This chapter fixes one set of conventions for the whole of the notes. They are the ones most common in the string-compactification literature, which are not always those used on the board; where a lecture writes a formula in another convention, it is converted to the one below.

Units and approximate equalities. We use natural units, $\hbar = c = 1$, so that masses, energies and inverse lengths are measured in GeV. An equality sign is exact. The sign "$\simeq$" means equality up to numerical factors of order one (such as 2, π or $\sqrt{4\pi}$) that are dropped when only the parametric dependence on g_s and $\mathcal{V}$ matters, and "$\sim$" means "of the order of". When a relation is first derived its factors are kept, and it is then quoted with "$\simeq$".

Metric and curvature. Metrics have mostly-plus signature, $\eta_{MN} = \mathrm{diag}(-1, +1, \ldots, +1)$, as in Polchinski [7], Tong [8], Wess and Bagger, and most of the compactification literature. The Riemann tensor is $R^{\rho}{}_{\sigma\mu\nu} = \partial_\mu \Gamma^{\rho}_{\nu\sigma} - \partial_\nu \Gamma^{\rho}_{\mu\sigma} + \Gamma^{\rho}_{\mu\lambda}\Gamma^{\lambda}_{\nu\sigma} - \Gamma^{\rho}_{\nu\lambda}\Gamma^{\lambda}_{\mu\sigma}$, the Ricci tensor is $R_{\mu\nu} = R^{\rho}{}_{\mu\rho\nu}$, and a sphere has positive Ricci scalar. With these choices the four-dimensional Einstein–Hilbert action is

$$S_{\mathrm{EH}} = \frac{M_P^2}{2} \int \mathrm{d}^4x \, \sqrt{-g}\, R .$$

For a Lorentzian metric we write $\sqrt{-g}$, for the Riemannian metric of the compact space $\sqrt{g}$.

Spaces, coordinates and indices. Ten-dimensional spacetime is $X_{10} = \mathbb{R}^{1,3} \times Y_6$, with Y_6 a compact six-dimensional manifold and coordinates $X^M = (x^\mu, y^m)$.

- Capital Latin indices $M, N, P = 0, \ldots, 9$ are ten-dimensional.
- Greek indices $\mu, \nu, \rho = 0, \ldots, 3$ are four-dimensional.
- Lower-case Latin indices $m, n, p = 4, \ldots, 9$ are real indices on Y_6.
- The indices $i, j = 1, 2, 3$ and $\bar{\imath}, \bar{\jmath}$ label holomorphic and antiholomorphic coordinates z^i, $\bar{z}^{\bar{\jmath}}$ on Y_6 when it is a complex manifold.
- The indices $a, b = 0, \ldots, p$ label the coordinates ξ^a on the worldvolume W_{p+1} of a Dp-brane, which is embedded in spacetime as $X^M = Y^M(\xi)$ (Section 4.3).

In Chapter 4 the indices i, j also label the directions transverse to a D-brane and the branes of a stack, and the index a of the worldvolume is unrelated to the label of the complex-structure moduli U_a; the context always tells which is meant. The ten-dimensional metric is G_{MN}, with determinant G and Ricci scalar R_{10}. On a product space it splits into the four-dimensional metric $g_{\mu\nu}$ (determinant g_4, Ricci scalar R_4) and the internal metric g_{mn} (determinant g_6, Ricci scalar R_6). The volume of the compact space is $\mathrm{Vol}(Y_6) = \int_{Y_6} \mathrm{d}^6y \, \sqrt{g_6}$. The holonomy group of Y_6 is $\mathrm{Hol}(Y_6)$, and $\mathbb{1}$ is the identity.

String length and string scale. The only dimensionful parameter of string theory is α', related to the string tension by $T = 1/(2\pi\alpha')$. Following the lecture and the moduli-stabilisation literature we define

$$\ell_s = 2\pi\sqrt{\alpha'}, \qquad M_s = \frac{1}{\ell_s}.$$

Polchinski and Tong use $\ell_s = \sqrt{\alpha'}$ instead, which is smaller by 2π. The virtue of the factor 2π is that it absorbs the powers of 2π of the ten-dimensional gravitational coupling, $2\kappa_{10}^2 = (2\pi)^7\alpha'^4 = \ell_s^8/2\pi$, and of the D$p$-brane tensions, $T_p = 2\pi M_s^{p+1}$ (Equation (4.9)); the physical tension of a Dp-brane in a vacuum is T_p/g_s (Equation (4.12)). The price is that T-duality, which exchanges a circle of radius R with one of radius α'/R, reads $R \to \alpha'/R = 1/(4\pi^2 M_s^2 R)$ (Equation (2.3)); the self-dual radius is $\sqrt{\alpha'} = \ell_s/2\pi$.

Dilaton, string coupling and frames. The dilaton ϕ is a field. Its vacuum expectation value is $\phi_0 \equiv \langle\phi\rangle$, and the string coupling is

$$g_s = e^{\phi_0}.$$

The *string frame* metric G_{MN} is the one to which the string couples on the worldsheet; in it the gravitational action carries a factor $e^{-2\phi}$ in front of the Ricci scalar (Equation (1.5)). The ten-dimensional *Einstein frame* metric is

$$G^E_{MN} = e^{-(\phi-\phi_0)/2}\, G_{MN},$$

in which the gravitational action takes the Einstein–Hilbert form $\frac{1}{2\kappa_{10}^2 g_s^2}\int \mathrm{d}^{10}X\sqrt{-G_E}\,R_{10}[G_E]$. Because ϕ_0 is subtracted, the two frames coincide in a vacuum with constant dilaton, $\phi = \phi_0$. In particular a volume measured in string units is the same number in both frames, and the relations between scales derived in Chapter 1, $M_P^2 = 4\pi\mathcal{V}M_s^2/g_s^2$ and $M_{KK} \simeq M_s/\mathcal{V}^{1/6}$, hold in either.

Much of the moduli-stabilisation literature uses instead $G^E_{MN} = e^{-\phi/2}G_{MN}$, without the subtraction. In a vacuum the two Einstein frames then differ by the constant factor $g_s^{-1/2}$, so that a dimensionless volume in that convention is $\mathcal{V}' = g_s^{-3/2}\,\mathcal{V}$, and the same relations read $M_s = g_s^{1/4}M_P/\sqrt{4\pi\mathcal{V}'}$ and $M_{KK} \simeq M_P/(\sqrt{4\pi}\,\mathcal{V}'^{2/3})$. Formulas taken from such references are converted with $\mathcal{V}' = g_s^{-3/2}\mathcal{V}$.

Volume. $\mathcal{V}$ is the dimensionless volume of Y_6 in units of ℓ_s^6, measured with the Einstein-frame metric:

$$\mathrm{Vol}(Y_6) = \mathcal{V}\,\ell_s^6.$$

Planck mass. M_P is the reduced Planck mass,

$$M_P = (8\pi G_N)^{-1/2} \simeq 2.4\times10^{18}\,\mathrm{GeV},$$

the coefficient of the four-dimensional Einstein–Hilbert action above. Four-dimensional actions are written in the four-dimensional Einstein frame, in which this coefficient is constant.

Counting supersymmetries. $\mathcal{N}$ counts supersymmetry generators in units of the smallest spinor of the spacetime dimension. In ten dimensions this is a Majorana–Weyl spinor, with 16 real components, so type IIA and IIB supergravity, with two of them, have ten-dimensional $\mathcal{N} = 2$ and 32 real supercharges. The two have opposite chirality in type IIA, and the same chirality in type IIB, which is therefore chiral. In four dimensions the unit is a Majorana spinor, equivalently a Weyl spinor and its conjugate, with 4 real components, so that four-dimensional $\mathcal{N} = 8, 4, 2, 1$ correspond to $32, 16, 8, 4$ real supercharges. A subscript $\pm$ on a ten-dimensional spinor, as in ψ_{M+} or λ_-, gives its chirality, the eigenvalue ± 1 of $\Gamma_{11} = \Gamma^0\Gamma^1\cdots\Gamma^9$, which squares to one in our signature.

Forms. A p-form $C_p = \frac{1}{p!} C_{M_1\ldots M_p}\, \mathrm{d}X^{M_1} \wedge \cdots \wedge \mathrm{d}X^{M_p}$ is identified with its totally antisymmetric components, e.g. $C_2 \leftrightarrow C_{MN} = C_{[MN]}$. Antisymmetrisation brackets have unit weight, $T_{[MN]} = \frac{1}{2}(T_{MN} - T_{NM})$ and in general $\frac{1}{p!}$ times the signed sum over the permutations of p indices, so that $\mathrm{d}C_p$ has components $(p+1)\,\partial_{[M_0} C_{M_1\ldots M_p]}$. For a one-form this is $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$. The NS–NS two-form is B_2, with components B_{MN} and field strength $H_3 = \mathrm{d}B_2$, i.e. $H_{MNP} = 3\,\partial_{[M} B_{NP]} = \partial_M B_{NP} + \partial_N B_{PM} + \partial_P B_{MN}$; its degree is always written, since H alone is the Hubble rate. The R–R forms are C_p, with field strengths $F_{p+1} = \mathrm{d}C_p + \ldots$. Forms, the exterior derivative and the Hodge star are introduced in Section 4.4. The Levi-Civita symbol has $\varepsilon_{01\cdots(D-1)} = +1$, with lower indices, on a D-dimensional Lorentzian manifold ($\varepsilon_{12\cdots D} = +1$ on a Riemannian one), and the Hodge star is defined by Equation (4.24), with $\sqrt{|g|}$ standing for $\sqrt{-g}$ or $\sqrt{g}$. On p-forms it obeys $\star\star = (-1)^{p(D-p)} s$, with $s = -1$ for a Lorentzian and $s = +1$ for a Riemannian metric, and $A_p \wedge \star B_p = \frac{1}{p!} A_{M_1\ldots M_p} B^{M_1\ldots M_p} \sqrt{|g|}\, \mathrm{d}^D x$ (Equation (4.25)), where $\mathrm{d}^D x$ is the wedge product of the coordinate differentials in increasing order. A form ω on Y_6 is harmonic if $\Delta\omega = 0$, with $\Delta = \mathrm{d}\mathrm{d}^\dagger + \mathrm{d}^\dagger\mathrm{d}$ the Laplacian of the internal metric. On a complex manifold a (p, q)-form has p holomorphic and q antiholomorphic indices.

Electromagnetism. Charges are in Heaviside–Lorentz units, so that $\alpha_{\mathrm{em}} = e^2/4\pi$, and the action of a $U(1)$ gauge field coupled to a current is $-\frac{1}{2}\int F_2 \wedge \star F_2 + \int A_1 \wedge \star J$, with $F_2 = \mathrm{d}A_1$ (Equation (4.26)). In mostly-plus signature $A_\mu = (-\Phi, \vec{A})$, $F_{0i} = -E_i$, $F_{ij} = \epsilon_{ijk} B_k$, and the electric and magnetic currents are $J^\mu = (\rho_e, \vec{J})$ and $K^\mu = (\rho_m, \vec{K})$. Maxwell's equations are then $\mathrm{d}\star F_2 = \star J$ and $\mathrm{d}F_2 = \star K$, i.e. $-\partial_\mu F^{\mu\nu} = J^\nu$ and $\partial_\mu (\star F)^{\mu\nu} = K^\nu$ (Equation (4.31)), with $\nabla\cdot\vec{E} = \rho_e$ and $\nabla\cdot\vec{B} = \rho_m$. In these units the Dirac quantisation condition reads $q_e q_m \in 2\pi\mathbb{Z}$ (Equation (4.33)).

Moduli. The Hodge numbers $h^{p,q}$ of Y_6 count its linearly independent harmonic (p, q)-forms; for a Calabi–Yau threefold the non-trivial ones are $h^{1,1}$ and $h^{2,1}$ ($= h^{1,2}$). A generic modulus is ϕ^I; in discussing the couplings of moduli to matter, ϕ without index also stands for a generic canonically normalised modulus. In Chapter 3 ϕ is likewise a single generic modulus, never the dilaton: in the toy models of the Dine–Seiberg problem a dimensionless one, $\mathrm{Re}\, S$ or $\mathcal{V}$, with critical points $\phi_{0,\pm}$, and in cosmology a canonically normalised one, measured from its late-time minimum, with mass m_ϕ and initial displacement ϕ_0. The Kähler moduli are T_i, $i = 1, \ldots, h^{1,1}$, and the complex-structure moduli U_a, $a = 1, \ldots, h^{2,1}$. The index i of T_i is a label, not a holomorphic index; the two are never needed in the same formula, and the context tells which is meant. The axio-dilaton is

$$S = e^{-\phi} + i\, C_0, \qquad \langle \mathrm{Re}\, S \rangle = \frac{1}{g_s},$$

as in the lecture and in the Large Volume Scenario literature. Much of the flux-compactification literature (GKP, KKLT, F-theory) uses instead $\tau = C_0 + i\,e^{-\phi} = i\bar{S}$, on which the $SL(2,\mathbb{Z})$ S-duality of type IIB acts by fractional linear transformations; formulas taken from there are converted with $\tau = i\bar{S}$. With an index, τ_i will denote the volumes of four-cycles, the real parts of the T_i. Gauge couplings are $\alpha = g^2/4\pi$; the hypercharge coupling is GUT-normalised, $\alpha_1 = \frac{5}{3}\alpha_Y$.

Cosmology. A spatially flat expanding universe has the Friedmann–Lemaître–Robertson–Walker metric $\mathrm{d}s^2 = -\mathrm{d}t^2 + a(t)^2\,\mathrm{d}\vec{x}^{\,2}$, with scale factor $a(t)$ and Hubble rate $H = \dot{a}/a$; dots are derivatives with respect to the cosmic time t. The Friedmann equation is $3H^2M_P^2 = \rho$, with ρ the total energy density. In an epoch dominated by radiation at temperature T, $\rho = (\pi^2/30)\,g_*T^4$, with g_* the number of relativistic degrees of freedom, so that $H \simeq T^2/M_P$. A subscript on H or T gives the event at which it is evaluated (H_{osc}, H_{dec}, T_{RH}, T_{BBN}), and Γ is a decay rate, the inverse of a lifetime.

Supergravity. Four-dimensional $\mathcal{N} = 1$ supergravity, with Kähler potential K and superpotential W, is written in the conventions of Wess and Bagger.

Chapter 1

From ten to four dimensions: scales and supersymmetry

Lecture of 5 November 2026

This first lecture is an overview of how one goes from ten-dimensional superstring theory to the four-dimensional world we observe; the following lectures fill in the details. Two questions organise it. The first is quantitative: when six of the ten dimensions are compact, which energy scales appear, and in which window of energies can we trust which effective description? The answer is a hierarchy of three scales, the Planck mass, the string scale and the Kaluza–Klein scale, controlled by two numbers, the string coupling g_s and the volume $\mathcal{V}$ of the extra dimensions in string units. The second question is qualitative: how much of the supersymmetry of ten-dimensional string theory survives in four dimensions? The answer depends on the holonomy of the compact space, and it is what singles out Calabi–Yau manifolds.

1.1 Where we stand and where we are going

The first part of the course, Prof. Ling Lin's module, described the string from the point of view of its worldsheet. The fundamental object is a vibrating string, and quantising it means quantising the two-dimensional field theory that lives on the surface it sweeps out, the worldsheet, whose fields are the spacetime coordinates X^M of the string. This is a two-dimensional conformal field theory. It has a Weyl symmetry, and the requirement that this symmetry has no anomaly fixes the dimension of spacetime, $D = 26$ for the bosonic string. The spectrum of the bosonic string contains a tachyon, a sign that the theory is unstable. To get rid of it one adds fermions on the worldsheet. The result is the superstring, whose critical dimension is $D = 10$. The Gliozzi–Scherk–Olive (GSO) projection removes the tachyon, and the theory turns out to be supersymmetric not only on the worldsheet but also in spacetime. The quantum states of the string form a tower of modes, a few of them massless and the others massive, and each of them corresponds to a spacetime field; among the massless modes of the closed string are the metric G_{MN}, the Kalb–Ramond two-form B_{MN} and the dilaton ϕ.

Besides closed strings there are open strings, whose endpoints are attached to extended objects, the Dp-branes, of any dimension up to that of spacetime. The open strings ending on a brane describe fields that live on it, in particular gauge fields, and on a stack of N coincident branes they give a non-abelian gauge theory with gauge group $U(N)$ (Section 4.2). There are five consistent perturbative superstring theories, which differ in how the spectrum is chosen: type I, types IIA and IIB, in which worldsheet fermions are added for both the left- and the

right-moving degrees of freedom of the closed string, and the two heterotic theories, with gauge groups $SO(32)$ and $E_8 \times E_8$, in which only the right-movers are supersymmetrised. Perturbatively they look different, but they are related by dualities, some of which exchange a strongly coupled theory with a weakly coupled one, and they are understood as different limits of a single underlying theory, M-theory (Section 2.2). Loop amplitudes are sums over Riemann surfaces of increasing genus, which replace the Feynman diagrams of field theory; since the string has a finite size, which acts as a natural cutoff, they have no ultraviolet divergences.

From here one could follow many directions: the quantum properties of each of the five theories, their mathematical structure, black holes in ten or fewer dimensions. This part of the course follows one of them, *string compactifications*: how to reproduce our four-dimensional world, with the Standard Model and cosmology, starting from ten-dimensional string theory. The idea is to take spacetime to be four-dimensional Minkowski space times a compact six-dimensional space, and to go to energies below the inverse size of the extra dimensions, where they can no longer be seen. What remains is a four-dimensional effective theory with local supersymmetry, a supergravity theory of the kind met in the supersymmetry course, with the important difference that it is not written down by hand: it is derived from a consistent theory of quantum gravity. Its physics depends strongly on the geometry and the topology of the extra dimensions, so we shall need some differential geometry and Calabi–Yau manifolds, and D-branes will play an essential role in reproducing the non-abelian gauge theories seen in nature. The module covers the low-energy effective action of string theory, D-brane actions and dualities, type II compactifications on Calabi–Yau manifolds, flux compactifications and moduli stabilisation, supersymmetry breaking in string models, and the basic principles of string cosmology.

We shall see that the way down from ten to four dimensions is not unique. There is a *landscape*, or a multiverse, of solutions, each corresponding to a different four-dimensional universe. The goal is to find at least one that reproduces what we know, the Standard Model and cosmology, and possibly also offers solutions to the open problems recalled at the beginning of the supersymmetry course: inflation, dark matter, dark energy, the hierarchy problem and supersymmetry breaking.

The subject is vast, and the aim here is a global understanding of the whole picture rather than a systematic derivation of one part of it. The first lectures are therefore largely qualitative, meant to build intuition; later lectures return to the main steps with explicit computations and the mathematical background they need (differential forms, complex manifolds, branes and the dualities that generalise electric–magnetic duality).

References. The following books, reviews and lecture notes cover the material of this part of the course, and much more:

1. L. E. Ibáñez and A. M. Uranga, *String Theory and Particle Physics: An Introduction to String Phenomenology* [1];

2. A. Hebecker, *Naturalness, String Landscape and Multiverse*, lecture notes freely available online [2];

3. R. Blumenhagen, B. Körs, D. Lüst and S. Stieberger, *Four-dimensional string compactifications with D-branes, orientifolds and fluxes*, a long review [3];

4. lecture notes by J. Louis, available on his website [4];

5. D. Baumann and L. McAllister, *Inflation and String Theory*, a book whose lecture-note version is available online [5];

6. M. Cicoli, J. P. Conlon, A. Maharana, S. Parameswaran, F. Quevedo and I. Zavala, *String cosmology: from the early universe to today* [6].

The first four are oriented towards particle physics and the low-energy limit of string theory. The last two are oriented towards cosmology, which is a particularly promising arena for testing string theory, because some cosmological questions, inflation above all, can only be addressed reliably in an ultraviolet-complete theory. The review [6] starts with an introduction to compactifications and then follows the whole cosmological history, from inflation through reheating to dark matter and dark energy.

1.2 Superstring theory as a theory of quantum gravity

Superstring theory is a consistent quantum theory, and the reason for the interest in it is that it is a consistent theory of quantum *gravity*. Its massless spectrum contains a symmetric tensor field $G_{MN}(X)$, the metric: there is a massless spin-two particle, the graviton, and so there is gravity, in a theory whose loop amplitudes are finite. But string theory is not only a theory of gravity. It also contains gauge interactions and matter, basically all the rest of physics, so that it is a candidate for a unified description of all interactions, and it is a framework large enough that other approaches to quantum gravity may turn out to be contained in it.

The theory is consistent because two potential problems are absent. There are no tachyons, thanks to the GSO projection, and there are no states of negative norm. The second requirement is the absence of a Weyl anomaly on the worldsheet, and it fixes the number of spacetime dimensions to the critical dimension

$$D = 10. \tag{1.1}$$

String theory is the only theory we know that predicts the number of spacetime dimensions. The prediction is, at first sight, wrong, since we see four dimensions. But it is not so bad: the answer is a positive integer, and it could have been worse.

1.3 Compactification and the Kaluza–Klein scale

To make contact with the real world we must study *string compactifications*. Ten-dimensional Minkowski space $\mathbb{R}^{1,9}$ is a solution of string theory, but there are others, in which ten-dimensional spacetime is the product of four-dimensional Minkowski space and a six-dimensional manifold,

$$X_{10} = \mathbb{R}^{1,3} \times Y_6, \tag{1.2}$$

where Y_6 is a *compact* space: a six-sphere, a six-dimensional torus, or something more complicated. Its volume $\mathrm{Vol}(Y_6)$ has dimensions of a length to the sixth power, so its $-1/6$ power is an energy, the *Kaluza–Klein scale*,

$$M_{KK} := \frac{1}{\mathrm{Vol}(Y_6)^{1/6}}. \tag{1.3}$$

M_{KK} is the energy above which one starts to see the extra dimensions; equivalently, $1/M_{KK}$ is the length scale below which they become visible. We expect

$$M_{KK} \gg E_{\mathrm{LHC}} \sim 1\,\mathrm{TeV},$$

the energies probed at the LHC, and this explains why we have not seen extra dimensions.

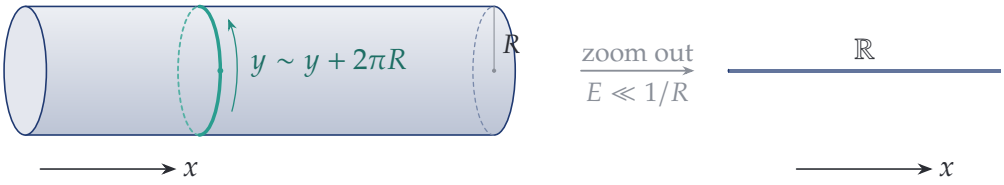


Figure 1.1: A cylinder $\mathbb{R} \times S^1$ seen from close by and from far away: at distances much larger than its circumference, or energies much lower than $1/R$, it looks like a line. Every point of the line carries a small circle, as every point of $\mathbb{R}^{1,3}$ carries a copy of Y_6.

The idea is easiest to picture in two dimensions. Take a spacetime that is a cylinder, with one extended direction and one direction curled up into a circle, and look at it from very far away: one sees a line, not the circle (Figure 1.1). In the same way, it is not so strange to expect that at every point of our four-dimensional spacetime there is a six-dimensional curled-up space that we do not see, simply because it is too small.

1.4 The string scale and the Planck scale

Let us now place the relevant scales on an energy axis. Somewhere high up is the four-dimensional Planck mass M_P; somewhere below it is the *string scale*, the inverse of the string length,

$$M_s = \frac{1}{\ell_s}, \qquad \ell_s = 2\pi\sqrt{\alpha'}. \tag{1.4}$$

The string length is set by the only free parameter of string theory, α', which determines the string tension, $T = 1/(2\pi\alpha')$. When the string is quantised, its vibrational modes give states that are either massless or have masses set by $1/\sqrt{\alpha'} = 2\pi M_s$, growing with the level of excitation. Why have we drawn M_s below M_P? To answer this, we have to relate the two.

1.4.1 The ten-dimensional action and the dilaton

From the scattering amplitudes of strings, at tree level and at loop level, one can reconstruct the effective action of the massless modes of the string in ten dimensions. Among its terms is the ten-dimensional Einstein–Hilbert term

$$S_{10} \supset \frac{1}{2\kappa_{10}^2} \int \mathrm{d}^{10}X \, \sqrt{-G} \, e^{-2\phi} \, R_{10}, \qquad \frac{1}{2\kappa_{10}^2} = \frac{2\pi}{\ell_s^8} = 2\pi \, M_s^8, \tag{1.5}$$

where R_{10} is the Ricci scalar built out of the ten-dimensional metric G_{MN}, and ϕ is the *dilaton*, another massless mode of the string. The factor $e^{-2\phi}$ multiplying the Ricci scalar means that this is the action in the *string frame* (see the conventions).

The remarkable property of the dilaton is that its vacuum expectation value fixes the strength of the interactions among strings, the *string coupling*,

$$g_s = e^{\langle\phi\rangle}. \tag{1.6}$$

The string coupling is therefore not a parameter that one can choose by hand, as one chooses α'. It is a function of the expectation value of a field: when the dilaton moves, the strength of the string interactions changes.

The action in Equation (1.5) is the tree-level contribution. It receives perturbative loop corrections and, on top of those, non-perturbative ones, whose size relative to the tree-level

action is controlled by g_s: for large g_s the tree-level action cannot be trusted, while in the limit $\langle\phi\rangle \to -\infty$ all the corrections become negligible. Computing all of them and finding the dynamics of the complete theory is not doable in practice. What one does instead is to *assume* that the string coupling is small, so that the tree-level action is a good first approximation, and then check a posteriori that the potential for ϕ has a minimum where g_s is indeed small. Since ϕ is determined dynamically, this is a consistency condition, not a choice: if it fails, the corrections should have been included from the start. How it can be met is the subject of Section 3.4. To trust perturbation theory we need

$$g_s \ll 1 \qquad \Longleftrightarrow \qquad \langle\phi\rangle \to -\infty. \tag{1.7}$$

1.4.2 Dimensional reduction and the Planck mass

We now perform the *dimensional reduction* of Equation (1.5) on the background of Equation (1.2), going to energies below the Kaluza–Klein scale. The ten-dimensional metric factorises into a four-dimensional piece and a six-dimensional one,

$$G_{MN}\,\mathrm{d}X^M \mathrm{d}X^N = g_{\mu\nu}(x)\,\mathrm{d}x^\mu \mathrm{d}x^\nu + g_{mn}(y)\,\mathrm{d}y^m \mathrm{d}y^n, \tag{1.8}$$

so that $\sqrt{-G} = \sqrt{-g_4}\,\sqrt{g_6}$ and $R_{10} = R_4 + R_6$. Keeping the term with the four-dimensional Ricci scalar, the action becomes

$$S_{10} \supset 2\pi M_s^8 \underbrace{\int_{Y_6} \mathrm{d}^6 y\,\sqrt{g_6}}_{\mathrm{Vol}(Y_6)} \int \mathrm{d}^4 x\,\sqrt{-g_4}\,e^{-2\phi}\,R_4. \tag{1.9}$$

There are other contributions, in particular the one with the internal Ricci scalar R_6, which we ignore for now. In the vacuum the dilaton takes its expectation value and can be taken outside the integral:

$$S_{10} \supset \underbrace{2\pi M_s^8\,\mathrm{Vol}(Y_6)\,e^{-2\langle\phi\rangle}}_{=M_P^2/2} \int \mathrm{d}^4 x\,\sqrt{-g_4}\,R_4. \tag{1.10}$$

This is the usual four-dimensional Einstein–Hilbert action, and its coefficient is the Planck mass, $M_P^2/2$. Writing the volume in terms of a dimensionless quantity $\mathcal{V}$, the volume in string units, $\mathrm{Vol}(Y_6) = \mathcal{V}\,\ell_s^6$, and using $e^{-2\langle\phi\rangle} = g_s^{-2}$ and $\ell_s^6 = M_s^{-6}$, we find

$$\boxed{M_P^2 = 4\pi\,\frac{\mathcal{V}}{g_s^2}\,M_s^2} \qquad \Longrightarrow \qquad M_s = \frac{g_s}{\sqrt{4\pi\mathcal{V}}}\,M_P \simeq g_s\,\frac{M_P}{\sqrt{\mathcal{V}}}. \tag{1.11}$$

Since ℓ_s is the only parameter of string theory, Equation (1.11) allows us to trade it for the Planck mass. Conceptually M_s is the fundamental scale and M_P a derived one, but M_P is the one we know, so in practice we fix $M_P \simeq 2.4 \times 10^{18}\,\mathrm{GeV}$ and derive everything else from it. The other two quantities in Equation (1.11) are not parameters at all: g_s is the expectation value of the dilaton, and we shall see that $\mathcal{V}$ is the expectation value of another field, the volume modulus. In the end everything is derived dynamically.

1.5 Perturbative control: string loops and α' corrections

Equation (1.11) tells us that

$$M_s \ll M_P \simeq 2.4 \times 10^{18}\,\mathrm{GeV} \qquad \text{for} \qquad g_s \ll 1 \quad \text{and} \quad \mathcal{V} \gg 1. \tag{1.12}$$

The first condition is that of Equation (1.7), needed to trust the string loop expansion. The second, that the volume be large in string units, is needed to trust another kind of perturbation theory. To see why, write the inverse volume as

$$\frac{1}{\mathcal{V}} = \frac{\ell_s^6}{\mathrm{Vol}(Y_6)} = \frac{(2\pi)^6\,\alpha'^3}{\mathrm{Vol}(Y_6)}. \tag{1.13}$$

The limit of large volume is the limit in which α' is small compared with the size of the extra dimensions, $\alpha'^3 \ll \mathrm{Vol}(Y_6)$. A string of fixed size moving in a volume that is huge compared with it is the same as a volume of fixed size in which the string length is sent to zero. But $\ell_s \to 0$ is the limit in which strings become point-like particles, and in which the Feynman diagrams of quantum field theory are recovered from the string diagrams. Corrections in $1/\mathcal{V}$ are therefore genuinely *stringy*: they tell us that the fundamental objects are strings, not point-like particles.

The effective theory thus receives two kinds of corrections, with two different expansion parameters. The first are *string loops*, the sums over Riemann surfaces of higher genus, with expansion parameter g_s; in the point-particle limit they reduce to the loop diagrams of quantum field theory. The second, with expansion parameter $1/\mathcal{V}$, are not loop corrections from the point of view of the spacetime field theory. They are tree-level corrections with higher derivatives, like the operators of dimension five, six and higher that integrating out heavy particles generates at tree level in field theory; here the heavy modes integrated out are the massive string modes. These α' *corrections* are higher-derivative terms in spacetime, and *worldsheet loops*, the loops of the two-dimensional field theory, counted by α', from the point of view of the worldsheet. In the regime $g_s \ll 1, \mathcal{V} \gg 1$ both string loops and α' corrections can be neglected, at least to a first approximation, and the effective theory can be trusted.

1.6 A hierarchy of scales and of effective theories

We can now complete the picture of the energy scales, shown in Figure 1.2. At the top is $M_P \simeq 2.4 \times 10^{18}\,\mathrm{GeV}$, and the string scale is below it, at a value we do not know, since it depends on g_s and $\mathcal{V}$ through Equation (1.11). At energies above M_s we see the whole ten-dimensional string theory: all the string resonances can be produced. The theory we shall study is type IIB string theory, which at present looks the most promising for obtaining phenomenology from string theory, for reasons that start to be explained in Chapter 2.

At energies below the string scale one remains in ten dimensions, but one can no longer produce the massive string modes, only the massless ones. What is left is no longer a string theory but a quantum field theory of point particles in ten dimensions, the fields that correspond to the massless modes of the string, with spectrum and interactions fixed by string theory. Since we started from a superstring this theory is supersymmetric, and since it contains gravity its supersymmetry is local: it is a ten-dimensional supergravity. For the type IIB string it is *type IIB supergravity*, with ten-dimensional $\mathcal{N} = 2$: there are two supersymmetry generators, two Majorana–Weyl spinors of the same chirality. "Type IIB" means that the field content is the massless spectrum of the type IIB string, not something chosen by hand.

Going to still lower energies we reach the Kaluza–Klein scale, below which the extra dimensions are invisible and the theory is four-dimensional. By Equations (1.3) and (1.11),

$$M_{KK} = \frac{1}{\mathrm{Vol}(Y_6)^{1/6}} = \frac{M_s}{\mathcal{V}^{1/6}} \simeq g_s\,\frac{M_P}{\mathcal{V}^{2/3}}, \tag{1.14}$$

where the power of the volume is $\frac{1}{6} + \frac{1}{2} = \frac{2}{3}$, and, keeping the factor of Equation (1.11), $M_{KK} = g_s M_P/(\sqrt{4\pi}\,\mathcal{V}^{2/3})$. At large volume the Kaluza–Klein scale is below the string scale,

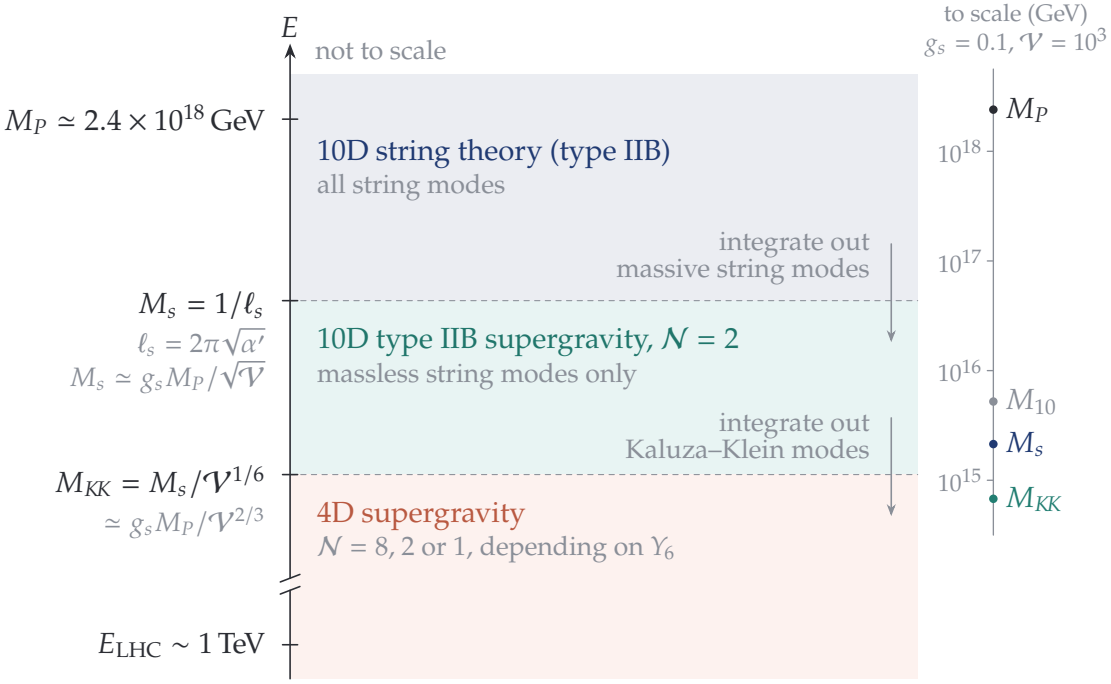


Figure 1.2: The hierarchy of scales of a string compactification and the effective description valid in each window of energies. The ratios are fixed by g_s and $\mathcal{V}$ through Equation (1.15).

$M_{KK}/M_s = \mathcal{V}^{-1/6} < 1$, so in the regime of Equation (1.12), in which we trust the effective theory, the hierarchy of scales is

$$M_{KK} < M_s < M_P, \qquad \frac{M_{KK}}{M_s} = \frac{1}{\mathcal{V}^{1/6}}, \qquad \frac{M_s}{M_P} = \frac{g_s}{\sqrt{4\pi\mathcal{V}}}. \tag{1.15}$$

Below M_{KK} one obtains a four-dimensional supergravity theory, described, as in global supersymmetry, by a Kähler potential and a superpotential. (In our conventions $\mathcal{V}$ in Equations (1.11) and (1.14) is also the Einstein-frame volume; the conventions give the same relations in the other common definition of the Einstein frame.)

1.7 How much supersymmetry survives: holonomy

The number of supersymmetries in four dimensions depends, of course, on the number of supersymmetries in ten dimensions, but it also depends on the topology of Y_6. The supersymmetry generators are spinors: in type IIB supergravity two ten-dimensional Majorana–Weyl spinors. Given one such spinor in ten dimensions, one can ask how many four-dimensional spinors it corresponds to, and the answer depends on the compact space. This is what singles out the famous Calabi–Yau manifolds. The statements in this section are made in words; we shall prove them later in this part of the course.

Consider first the simplest situation, a flat internal manifold, the six-dimensional torus T^6. In this case the largest amount of supersymmetry is preserved: one Majorana–Weyl spinor in ten dimensions corresponds to four spinors in four dimensions, and the two of type IIB give four-dimensional $\mathcal{N} = 8$ supergravity,

$$Y_6 = T^6, \quad \mathrm{Hol}(T^6) = \{\mathbb{1}\} \qquad \Longrightarrow \qquad \text{4D } \mathcal{N} = 8. \tag{1.16}$$

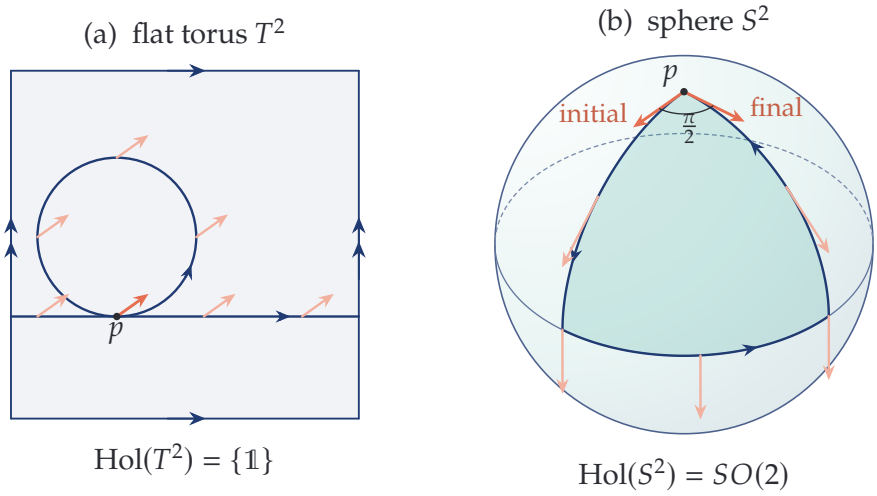


Figure 1.3: Parallel transport around closed loops. On a flat torus a vector always returns to itself, so the holonomy is trivial. On a sphere a vector carried around a closed loop returns rotated, here by $\pi/2$; the holonomy group collects all such rotations.

The property of T^6 that matters here is that it is a space of trivial *holonomy*: its holonomy group contains only the identity.

The holonomy is a simple notion. Take a manifold, choose a closed path in it starting and ending at a point p, take a vector at p and parallel-transport it around the path, as in Figure 1.3. When it gets back to p the vector has the same length, but it need not be the same vector. On a flat manifold it always comes back to itself, but on a curved manifold it is in general rotated. A different closed path gives in general a different rotation. The *holonomy group* is the set of all the transformations that relate the original vector to the transported one, for all possible closed paths. If the holonomy group contains only the identity, the vector always comes back to itself, as in the flat case; this is the trivial situation of the torus, which is flat.

This result is a nice consistency check. In the supersymmetry course we saw that $\mathcal{N} = 8$ is the largest amount of supersymmetry a four-dimensional theory can have: beyond it, the supermultiplets contain massless particles of spin higher than two, which have no consistent interactions. String theory never gives more than $\mathcal{N} = 8$, as a consistent theory must. Type IIB string theory on $\mathbb{R}^{1,3} \times T^6$ is a perfectly good solution, but it does not describe our world, because $\mathcal{N} = 8$ is too much supersymmetry. The world needs at most $\mathcal{N} = 1$ at high energies: in four-dimensional theories with $\mathcal{N} \geq 2$ the fermions cannot be chiral, while those of the Standard Model are. (Chirality in ten dimensions is a different matter: the chiral type IIB theory gives the non-chiral $\mathcal{N} = 8$ theory on T^6, and the non-chiral type IIA theory can give chiral four-dimensional theories when compactified down to $\mathcal{N} = 1$.) The supersymmetry must be reduced, and the way to reduce it is what selects Calabi–Yau manifolds.

If instead of T^6 the space Y_6 is a *Calabi–Yau manifold*, whose holonomy group is $SU(3)$, then one Majorana–Weyl spinor in ten dimensions gives a single spinor in four dimensions, and type IIB gives four-dimensional $\mathcal{N} = 2$ supergravity:

$$Y_6 = \text{Calabi–Yau}, \quad \text{Hol}(Y_6) = SU(3) \qquad \Longrightarrow \qquad \text{4D } \mathcal{N} = 2. \tag{1.17}$$

This is why Calabi–Yau manifolds, the six-dimensional manifolds of $SU(3)$ holonomy, became famous: they break the largest possible amount of supersymmetry while preserving some of it, which is what is needed to make contact with the real world.

To reach four-dimensional $\mathcal{N} = 1$ one more step is needed. Half of the degrees of freedom have to be projected away: one has to cut the spectrum. The operation that does it in type IIB is the *orientifold projection*. It reverses the orientation of the strings and identifies different modes of the string with each other, and keeping only the combinations invariant under it projects out half of the spectrum, and with it half of the supersymmetries. Type IIB string theory on a Calabi–Yau orientifold gives four-dimensional $\mathcal{N} = 1$ supergravity. We shall understand later in more detail what this operation means.

All these statements hold at the compactification scale, which is not known but is expected to be very high, perhaps 10^{15}–10^{18} GeV. There, depending on the internal manifold, one has four-dimensional $\mathcal{N} = 8$ (T^6), $\mathcal{N} = 2$ (a Calabi–Yau) or $\mathcal{N} = 1$ (a Calabi–Yau orientifold). The remaining task is to break the last $\mathcal{N} = 1$ to $\mathcal{N} = 0$ dynamically, at low energy, around the TeV scale.

The plan of this part of the course is to go through these steps, from ten-dimensional type IIB supergravity down to four-dimensional theories with $\mathcal{N} = 8$, $\mathcal{N} = 2$ and finally $\mathcal{N} = 1$, together with the geometric notions, such as holonomy, that they require. Note that the background $\mathbb{R}^{1,3} \times Y_6$ is put in by hand, together with its allowed deformations: Y_6 can be a torus or a Calabi–Yau manifold, and the Calabi–Yau can change in size or in shape. (A change of shape at fixed size is like deforming a doughnut into a teacup: there is still one hole, and the volume may stay the same, but the shape changes.) These deformations will be the moduli of the compactification (Section 2.7). Understanding dynamically why we end up in this background rather than in another, for example $\mathbb{R}^{1,4} \times Y_5$, is beyond present reach: it requires the full non-perturbative M-theory, not perturbative string theory.

Chapter 2

String phenomenology: from the heterotic string to the moduli problem

Lecture of 6 November 2026

This lecture continues the general description, begun in Chapter 1, of how string theory can be connected with the real world; like the first, it is mostly qualitative. It has three parts. The first explains why we look for compactifications that preserve four-dimensional $\mathcal{N} = 1$ supersymmetry at the Kaluza–Klein scale. The second surveys the five superstring theories in ten dimensions, the dualities that relate them and their massless spectra, and uses them to explain why string phenomenology started with the heterotic string and why it later moved to type II theories. The third introduces a problem that every string compactification has to face, and around which much of this course is organised: the *moduli*, massless scalar fields that parametrise the size and the shape of the extra dimensions, and the need to stabilise them.

2.1 Why four-dimensional $\mathcal{N} = 1$ supersymmetry

In the chain of effective theories of Section 1.6 (Figure 1.2), type IIB string theory gives type IIB supergravity below the string scale, and a four-dimensional supergravity below the Kaluza–Klein scale, once the Kaluza–Klein towers (Section 1.3) are integrated out; how many supersymmetries survive depends on the topology of Y_6 (Section 1.7). We pick Calabi–Yau manifolds, to be defined properly later, because they break the largest amount of supersymmetry while preserving some of it, and, after the orientifold projection, lead to $\mathcal{N} = 1$ at the Kaluza–Klein scale.

One could also consider compactifications that break supersymmetry completely, with $\mathcal{N} = 0$ at the Kaluza–Klein scale, and do model building directly there. We do not, for four reasons.

The first is that $\mathcal{N} = 1$ is the only chiral version of four-dimensional supersymmetry: in theories with $\mathcal{N} \geq 2$ the fermions come in non-chiral representations (Section 1.7). With $\mathcal{N} = 1$ there is hope of obtaining the Standard Model, which is a chiral theory.

The second concerns the scale of supersymmetry breaking. The Kaluza–Klein scale is not known: it could be close to the TeV scale, at the edge of detection, but more realistically it is very high, say 10^{14}–10^{15} GeV. The $\mathcal{N} = 1$ supersymmetry that survives there should then be broken spontaneously at much lower energies, for example at the TeV scale, to address the hierarchy problem, and *dynamically*: the hope is that non-perturbative effects, as in confinement, generate a scale exponentially smaller than the fundamental ones. Supersymmetry broken at low energies has the good properties met in the supersymmetry

course: it addresses the hierarchy problem, provides a dark matter candidate, leads to radiative electroweak symmetry breaking, in which loop corrections break the electroweak symmetry dynamically, and makes the gauge couplings unify. The hope is to recover all of these from a top-down approach. Whether this works is a property of the effective theory: one has to derive it from string theory, compute the non-perturbative effects, and check whether they break the remaining $\mathcal{N} = 1$ at low energies. It might also turn out that this is never possible.

The third concerns how the breaking is communicated. Besides knowing whether supersymmetry is broken at low energies, one must understand how the breaking is mediated to the *visible sector*, the supersymmetric version of the Standard Model, for instance the minimal supersymmetric Standard Model (MSSM). Some other sector of the compactification is responsible for breaking supersymmetry, and it must couple to the MSSM in such a way that the MSSM feels the breaking and the supersymmetric partners become massive. The breaking felt by the MSSM must be *soft*, so that the cancellation of quadratic divergences survives, and soft supersymmetry breaking is generated naturally in supergravity theories derived from string theory, through gravitational interactions. This is *gravity mediation*.

These three reasons are phenomenological. The fourth is theoretical, and we have not met it yet in the supersymmetry course: $\mathcal{N} = 1$ *supersymmetry allows one to control the quantum corrections to the effective theory.* A supersymmetric theory is specified by its Kähler potential K and its superpotential W, and one can ask how these quantities behave under quantum corrections. Each of them has a tree-level part, loop corrections and non-perturbative corrections, but not on the same footing:

$$K = K_{\text{tree}} + K_{\text{loops}} + K_{\text{np}}, \qquad W = W_{\text{tree}} + W_{\text{np}}. \tag{2.1}$$

The superpotential is exact at tree level in perturbation theory: as a consequence of supersymmetry and of holomorphy it receives no loop corrections, only non-perturbative ones (this will be proved in the supersymmetry course). This is a very powerful result: once W is computed at tree level, one knows for sure that there are no loop corrections. The Kähler potential, instead, does receive loop corrections. Since computing quantum corrections in string theory is extremely hard, this control is very important, and in a compactification that breaks supersymmetry completely it is absent. Keeping some supersymmetry helps to compute the effective theory, but not too much of it: with $\mathcal{N} = 1$, and not $\mathcal{N} = 2$, the theory remains chiral and good for phenomenology. That is the balance.

In an $\mathcal{N} = 0$ compactification a ten-dimensional Majorana–Weyl spinor corresponds to no spinor at all in four dimensions, so a ten-dimensional supersymmetry generator gives nothing. An example is $Y_6 = S^6$, a six-dimensional sphere, whose holonomy group is $SU(4)$ in the lecture's words, to be compared with $SU(3)$ for a Calabi–Yau manifold. Why the holonomy group determines the number of four-dimensional supersymmetries will be explained towards the end of the course.

2.2 Five superstring theories and their dualities

Our starting point will be ten-dimensional type IIB string theory, because, as we shall try to explain, it is at present the most promising one for phenomenology. In the end the starting point does not matter much, since all string theories are related by dualities. What changes from one description to another is which quantities are easy to compute: something that looks weakly coupled, and so computable, in one description can look strongly coupled, and hard to compute, in another.

The five consistent perturbative superstring theories recalled in Section 1.1, type IIA, type IIB, type I and the two heterotic theories, with gauge groups $SO(32)$ and $E_8 \times E_8$, are all

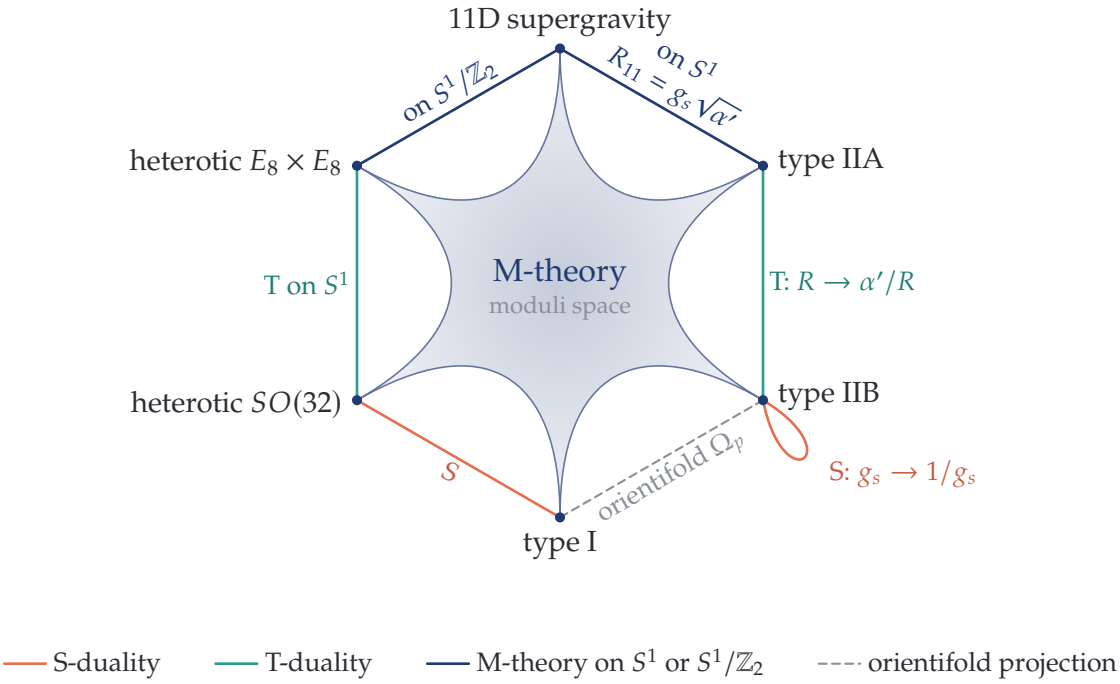


Figure 2.1: The five ten-dimensional superstring theories and eleven-dimensional supergravity as limits of a single theory, M-theory, and the dualities that relate them.

related by dualities of two kinds. *S-dualities* are strong–weak coupling dualities: they act on the dilaton, and exchange a theory at string coupling g_s with one at coupling $1/g_s$. *T-dualities* act instead on the geometry: they change the radius of the compactification manifold. The five theories are therefore best thought of not as different theories but as different limits, or better different perturbative corners, of a single, more fundamental theory, *M-theory*. What we know of M-theory is its low-energy limit, eleven-dimensional supergravity. In it, the size of the eleventh dimension corresponds to the string coupling. In a perturbative corner, where the coupling goes to zero and perturbation theory can be used, the size of the eleventh dimension goes to zero, and one recovers a ten-dimensional theory; when the coupling becomes large, one starts to see the eleventh dimension.

The simplest example is the relation between M-theory and type IIA. Compactified on a circle S^1 of radius R, going from eleven to ten dimensions, M-theory gives type IIA string theory, and R is set by the dilaton in such a way that

$$R \to 0 \qquad \Longleftrightarrow \qquad g_s \to 0 \qquad \text{(type IIA limit)}. \tag{2.2}$$

Type IIA is in turn T-dual to type IIB. T-duality changes the size of the compactification manifold: type IIA compactified on a circle of radius R is the same theory as type IIB compactified on a circle of radius

$$\tilde{R} = \frac{\alpha'}{R} = \frac{1}{4\pi^2 M_s^2 R}. \tag{2.3}$$

The two are the same theory, with R exchanged for α'/R: an example of a duality. The whole web of dualities is shown in Figure 2.1.

Sector	Type IIA	Type IIB	Heterotic
NS–NS	G_{MN}, B_{MN}, ϕ	G_{MN}, B_{MN}, ϕ	G_{MN}, B_{MN}, ϕ, A^a_M
R–R	C_1, C_3	C_0, C_2, C_4	—
NS–R	ψ^1_{M+}, λ^1_-	ψ^1_{M+}, λ^1_-	ψ_{M+}, λ_-, λ^a_+
R–NS	ψ^2_{M-}, λ^2_+	ψ^2_{M+}, λ^2_-	—
10D supersymmetry	$\mathcal{N} = 2$, non-chiral	$\mathcal{N} = 2$, chiral	$\mathcal{N} = 1$, chiral

Table 2.1: Massless spectra of the type IIA, type IIB and heterotic superstrings in ten dimensions, sorted by sector: the first label refers to the left-movers, the second to the right-movers (for the heterotic string see the text). G_{MN} is the metric, B_{MN} the Kalb–Ramond two-form, ϕ the dilaton, C_p the R–R p-forms, ψ_M the gravitini, λ the dilatini; A^a_M and λ^a are the heterotic gauge bosons and gaugini, $a = 1, \ldots, \dim G = 496$, with $G = SO(32)$ or $E_8 \times E_8$. The subscripts ± give the ten-dimensional chirality.

2.3 Massless spectra and the first string models

String phenomenology, the attempt to obtain the real world from string theory, started in the mid-1980s: string theory was born at the end of the 1960s and developed in the 1970s, and in the mid-1980s people started trying to connect it with the real world.

These first attempts used not type IIB but the $E_8 \times E_8$ heterotic string. The type II theories seemed to give no non-abelian gauge bosons in four dimensions, which are needed for the Standard Model, and so, although consistent, they seemed unfit for the real world. The heterotic string, instead, has a gauge group in its massless spectrum, $E_8 \times E_8$, and E_8 is a huge group which contains the Standard Model gauge group. The idea was to break one E_8 factor down to the Standard Model with a Higgs mechanism, and to use the dynamics of the other to break supersymmetry. This programme turned out to be very hard, for reasons explained below, although heterotic phenomenology is still pursued today. Around 1995 came the discovery of *D-branes* as dynamical objects of string theory, and with it the realisation that the open strings ending on them can behave as non-abelian gauge bosons (Section 4.2). D-branes are present in type II theories, so one had simply missed an ingredient: the non-abelian gauge bosons are absent from the closed-string spectrum, but they can come from open strings. Since then type II theories have been studied more than the heterotic string for phenomenology.

To see why, we list the massless modes of the closed string and the ten-dimensional fields they correspond to, in Table 2.1, for type IIA, type IIB and the heterotic string.

The labels NS and R stand for Neveu–Schwarz and Ramond, and they are boundary conditions for the fermions on the worldsheet. The bosonic coordinates $X^M(\sigma)$ of a closed string must come back to themselves when one goes around the string, $\sigma \to \sigma + 2\pi$. When the theory is supersymmetrised one must add worldsheet fermions ψ^M, and these can come back either to themselves or to minus themselves:

$$\psi^M(\sigma + 2\pi) = +\psi^M(\sigma) \quad \text{(Ramond)}, \qquad \psi^M(\sigma + 2\pi) = -\psi^M(\sigma) \quad \text{(Neveu–Schwarz)}. \tag{2.4}$$

On a closed string there are left-movers and right-movers, and the condition can be chosen independently for each. In type II theories there are worldsheet fermions for both, and so two spacetime supersymmetries, one coming from the left-movers and one from the right-movers; this is why they are called type II, and why all four combinations NS–NS, R–R, NS–R and R–NS appear. In the heterotic string there are worldsheet fermions only for the right-movers,

and so only one spacetime supersymmetry. The left-movers have no worldsheet fermions at all: they are those of the bosonic string. Being bosonic, they produce only spacetime bosons, as an NS sector does, which is why the two heterotic sectors are labelled NS–NS and NS–R, according to the boundary condition of the right-movers.

Now the fields. The NS–NS sector always contains the metric G_{MN}, the antisymmetric Kalb–Ramond two-form B_{MN} and the dilaton ϕ, the spectrum of any bosonic string theory. The heterotic string has in addition gauge bosons A^a_M, ten-dimensional vectors (hence the index M, not μ), and many of them: $a = 1, \ldots, \dim G$, with $G = SO(32)$ or $G = E_8 \times E_8$, both of dimension 496 ($\dim E_8 = 248$). (E_6, E_7 and E_8 are the exceptional Lie groups, related to the octonions.) This is why people expected the heterotic string to work: with non-abelian gauge bosons in ten dimensions, the four-dimensional theory would inherit some of them.

In the R–R sector type IIA has a one-form C_1 and a three-form C_3, and type IIB a zero-form C_0, a two-form C_2 and a four-form C_4. These are differential forms, i.e. antisymmetric tensors, such as $C_2 \leftrightarrow C_{MN} = C_{[MN]}$ and $C_4 \leftrightarrow C_{MNPQ} = C_{[MNPQ]}$, and C_0 is a pseudoscalar. They are all bosons. There are also higher forms, but they are dual to the ones listed: a p-form C_p in ten dimensions is equivalent to an $(8-p)$-form, since the field strengths are related by $F_{p+1} = \star F_{9-p}$, and in type IIB the field strength of C_4 is self-dual, $F_5 = \star F_5$.

The NS–R and R–NS sectors give fermions. There are two *gravitini* ψ_M, of spin 3/2, since there are as many gravitini as supersymmetries and type II theories have ten-dimensional $\mathcal{N} = 2$, and two *dilatini* λ, of spin 1/2, the supersymmetric partners of the dilaton. The subscripts ± give the ten-dimensional chirality, the eigenvalue of Γ_{11}, the ten-dimensional analogue of γ_5. In type IIA the two gravitini have opposite chiralities, and so do the two dilatini: type IIA is a non-chiral theory in ten dimensions. In type IIB the two gravitini have the same chirality, and so do the two dilatini: type IIB is a chiral theory in ten dimensions (see also the conventions). The heterotic string has a single supersymmetry in ten dimensions, hence one gravitino and one dilatino, and in addition the supersymmetric partners of the gauge bosons, the *gaugini* λ^a: it is ten-dimensional $\mathcal{N} = 1$ supergravity coupled to super-Yang–Mills theory with gauge group G.

2.4 Gauge bosons in four dimensions

The ten-dimensional gauge bosons of the heterotic string lead, in four dimensions, to four-dimensional gauge bosons: choosing the index $M = \mu$ and keeping the zero mode, constant along Y_6, one obtains $A^a_\mu(x)$, non-abelian gauge bosons that can give the gluons and the W and Z bosons of the Standard Model. In type IIA and type IIB this is not possible, and the reason is worth understanding, because it will come back in the discussion of moduli.

A ten-dimensional two-form, an object with two indices, could give objects with one index in four dimensions. Taking for instance $M = \mu$ along $\mathbb{R}^{1,3}$ and $N = n$ along Y_6, the component $B_{\mu n}$ is, from the four-dimensional point of view, a vector labelled by n. To extract its massless modes one expands it in the zero modes of the internal Laplacian, which for an object with one internal index are the harmonic one-forms $\omega^I_n(y)$ of Y_6:

$$B_{\mu n}(x, y) = \sum_I A^I_\mu(x)\, \omega^I_n(y) + (\text{massive Kaluza–Klein modes}). \tag{2.5}$$

One is effectively reducing B on one-cycles, one-dimensional submanifolds of the Calabi–Yau. But a Calabi–Yau manifold has no harmonic one-forms, $b_1(Y_6) = 0$, so these components vanish, and neither G_{MN} nor B_{MN} can give a vector $A_\mu(x)$ in four dimensions. We shall see later why this is a property of the Calabi–Yau.

The R–R forms offer more possibilities. C_1 in type IIA is already a one-form, and gives a vector directly. A form with more indices gives a vector if it is reduced on a cycle with one

dimension less than its degree: C_3 in type IIA reduced on a two-cycle ($3-2=1$), or C_4 in type IIB reduced on a three-cycle ($4-3=1$). In principle all of these behave in four dimensions as objects with one index. This does not work either, for a reason that is a property of the effective theory: the R–R forms do not couple to any massless mode of the string. The photon of the Standard Model must couple to leptons and quarks, while the R–R photons couple only to objects with masses at or above the string scale, the D-branes that we shall meet in Chapter 4; moreover they are abelian. The closed-string spectrum of type II theories does give photons, but not the photon of the Standard Model.

This is an interesting prediction of string theory: on top of the ordinary photon there are many other massless abelian gauge bosons, *dark photons*, dark because they do not couple to ordinary matter except gravitationally. One way to detect them uses cosmology. If they were produced in the early universe, they behave in practice like extra neutrinos: they have spin one rather than one half, but they carry no charge, are relativistic, since they are massless, and are effectively decoupled from the Standard Model. As a dark relativistic species they add to the energy density of the universe and modify its expansion rate, so measuring the expansion rate either reveals something beyond the three neutrinos or strongly constrains these models.

2.5 Heterotic model building

For these reasons, in the mid-1980s people focused on the $E_8 \times E_8$ heterotic string. One E_8 factor is the *visible sector*: a symmetry breaking (a Higgs mechanism) inside it brings it down to a grand unified theory (GUT). The relevant subgroups are

$$E_8 \supset SO(10) \supset SU(5) \supset SU(3) \times SU(2) \times U(1). \tag{2.6}$$

$SO(10)$ and $SU(5)$ are the groups used for grand unification, and a Higgs mechanism, similar to the one that breaks $SU(2)_L \times U(1)_Y$ in the Standard Model, breaks $SU(5)$ down to $SU(3) \times SU(2) \times U(1)$. At the beginning there is a single group factor, so a single coupling: a unified theory with only one non-gravitational interaction, which then breaks down to the three interactions of the Standard Model.

The second E_8 factor is a *hidden sector*: the leptons and quarks of the Standard Model are not charged under it, and the matter charged under it is not charged under the Standard Model. The hidden sector is responsible for supersymmetry breaking, through a dynamics very similar to confinement in QCD. In QCD the gauge coupling grows towards low energies and blows up, and quarks and antiquarks condense: the quark bilinear develops a non-zero vacuum expectation value, $\langle q\bar{q}\rangle \neq 0$. Similarly, strong dynamics at low energies in the hidden E_8 gives a non-zero vacuum expectation value to some fields and breaks supersymmetry. Gravitational interactions, which are always present, then mediate the breaking to the visible sector, which becomes a non-supersymmetric Standard Model. That was the idea.

2.6 The problems of the heterotic string: the GUT coupling

The heterotic string, however, has problems that have not yet been fully addressed. The first concerns the gauge coupling. In a heterotic compactification that leads to a grand unified theory, based on $SO(10)$ or $SU(5)$, the GUT coupling is controlled by the string coupling and by the volume of the Calabi–Yau manifold in string units (for now, take it for granted):

$$\alpha_{\rm GUT} \equiv \frac{g^2_{\rm GUT}}{4\pi} \simeq \frac{g_s^2}{\mathcal{V}}. \tag{2.7}$$

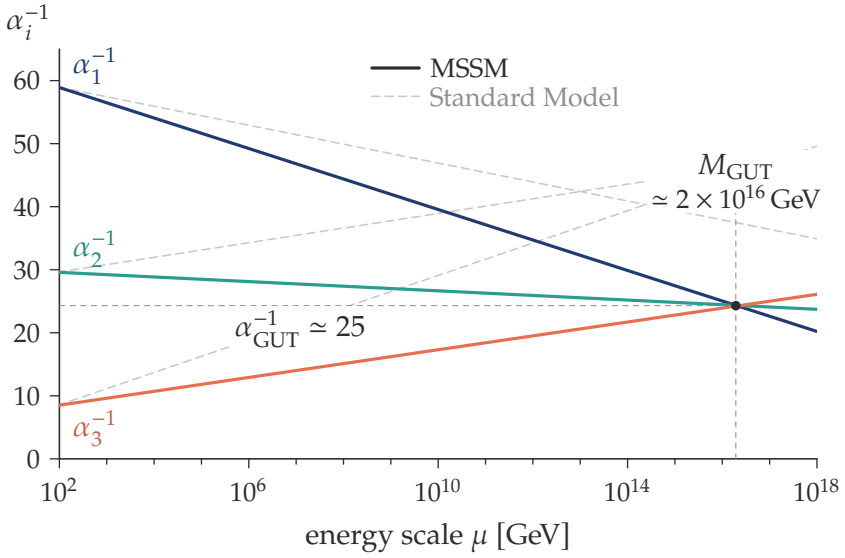


Figure 2.2: Running of the three inverse gauge couplings of the Standard Model gauge group with the MSSM spectrum, at one loop. They meet at $\alpha_{\text{GUT}}^{-1} \simeq 25$ at $M_{\text{GUT}} \simeq 2 \times 10^{16}$ GeV.

From phenomenology, on the other hand, we know the value of α_{GUT}: the three gauge couplings of the Standard Model, run with the renormalisation group of the MSSM, tend to unify at

$$\alpha_{\text{GUT}}^{-1} \simeq 25, \tag{2.8}$$

as shown in Figure 2.2.

But then Equation (2.7) requires

$$\mathcal{V} \simeq 25\, g_s^2, \tag{2.9}$$

so one cannot have at the same time $g_s \ll 1$ and $\mathcal{V} \gg 1$, the two conditions of Equation (1.12) for trusting the effective theory. At small string coupling the volume is of order one or smaller, and the higher-derivative (α') corrections are very large; at large volume, where they are suppressed, the string coupling is large, and the loop corrections can never be ignored. Either way it is hard to trust the effective theory. We shall see that type II theories avoid this problem. There the gauge fields live on D-branes, and their coupling is set by the size of the cycle that the branes wrap, not by $\mathcal{V}$, so one can go to large volume and small string coupling while keeping $\alpha_{\text{GUT}}^{-1} \simeq 25$.

2.7 Moduli

The second problem introduces a concept that applies to every string theory, not only to the heterotic string: *moduli stabilisation*. Consider the dimensional reduction of the ten-dimensional metric to four dimensions. Starting from one metric in ten dimensions, one obtains many scalars in four dimensions, the *moduli*. On $X_{10} = \mathbb{R}^{1,3} \times Y_6$ (Equation (1.2)), with coordinates x^μ on $\mathbb{R}^{1,3}$ and y^m on the Calabi–Yau manifold Y_6, the metric $G_{MN}(x, y)$ splits into three kinds of components:

$$G_{MN}(x,y) \longrightarrow \begin{cases} G_{\mu\nu}(x), \\ G_{\mu m}(x,y) = \sum_I A_\mu^I(x)\, \omega_m^I(y) = 0 \quad \text{for a Calabi–Yau}, \\ G_{mn}(x,y) = \sum_I \phi^I(x)\, \omega_{mn}^I(y). \end{cases} \tag{2.10}$$

The first, with both indices along $\mathbb{R}^{1,3}$, gives the four-dimensional metric. The mixed components $G_{\mu m}$ are four-dimensional vectors, whose zero modes would come from the harmonic one-forms $\omega^I_m(y)$ of Y_6, exactly as for $B_{\mu n}$ in Equation (2.5). A Calabi–Yau has none ($b_1 = 0$, Section 2.4), so the metric gives no vectors.

The internal components G_{mn} are four-dimensional scalars, since m and n are just labels. They are expanded in scalar fields $\phi^I(x)$ times objects $\omega^I_{mn}(y)$ with two internal indices, and the number of terms is a topological property of Y_6: in practice I takes hundreds of values. A metric in ten dimensions therefore gives a metric in four dimensions plus many scalars, which parametrise the internal metric: moving in the space of these fields changes the size or the shape of the Calabi–Yau, as anticipated at the end of Section 1.7. Mathematicians call such parameters of the Calabi–Yau metric moduli; in physics they are scalar fields. One of them is the volume $\mathcal{V}$ that appears in α_{GUT}, Equation (2.7). As a field, $\mathcal{V}(x)$ is a function on Minkowski space: the Calabi–Yau attached at the point x can have a different volume from the one attached at another point.

There are, in general, three types of moduli.

Kähler moduli. The first type parametrises deformations of the size of Y_6; an example is the volume. They are denoted

$$T_i(x), \qquad i = 1, \ldots, h^{1,1}(Y_6), \tag{2.11}$$

where $h^{1,1}$, a *Hodge number*, is the number of linearly independent harmonic $(1,1)$-forms on Y_6, those annihilated by the Laplacian, $\Delta\omega = 0$: a topological property. The name $(1,1)$ comes from the fact that Y_6 is a complex manifold, on which one can use holomorphic and antiholomorphic coordinates z^i and $\bar{z}^{\bar{\imath}}$, as one describes the plane either with (x, y) or, as the complex plane, with $(z, \bar{z})$. A $(1,1)$-form has one holomorphic and one antiholomorphic index, $\omega_{i\bar{\jmath}}\, \mathrm{d}z^i \wedge \mathrm{d}\bar{z}^{\bar{\jmath}}$.

Complex-structure moduli. The second type changes the complex structure of Y_6, i.e. how complex coordinates are assigned to it. Without changing the volume, these deformations change the shape of Y_6. They are denoted

$$U_a(x), \qquad a = 1, \ldots, h^{2,1}(Y_6), \tag{2.12}$$

where the Hodge number $h^{2,1}$ counts the linearly independent harmonic $(2,1)$-forms on Y_6. The beauty of the construction is that topological properties of Y_6 correspond to new scalar fields in four dimensions.

The axio-dilaton. The third type comes not from the metric but from the other scalars of the massless spectrum of type IIB, Table 2.1: the dilaton ϕ and the R–R zero-form C_0, a scalar since it has no index. They combine into a complex scalar, the *axio-dilaton*

$$S = e^{-\phi} + i\, C_0. \tag{2.13}$$

All the moduli turn out to be complex: T_i and U_a, like S, are complex scalar fields. The real part of S is $e^{-\phi}$, so by Equation (1.6) its expectation value is the inverse string coupling,

$$\langle \mathrm{Re}\, S \rangle = \frac{1}{g_s} \gg 1, \tag{2.14}$$

which must be large for the effective theory to be trusted.

For a typical Calabi–Yau manifold both $h^{1,1}$ and $h^{2,1}$ can be of order 100, so that one metric in ten dimensions gives hundreds of scalars in four. Calabi–Yau manifolds are named after Eugenio Calabi, an Italian-born mathematician, who conjectured their existence, and Shing-Tung Yau, a Chinese-born mathematician, who proved it; mathematicians and

theoretical physicists then started building explicit examples. Their number is still not known. The largest dataset built so far, that of Kreuzer and Skarke [9], has about half a billion examples, 473 800 776 (it lists the four-dimensional reflexive polytopes, each of which defines Calabi–Yau threefolds as hypersurfaces in a toric variety), each a good string compactification. A few have both Hodge numbers small, for instance $h^{1,1} = 1$ and a small $h^{2,1}$, and so only a few scalars, but most have many; for the compactification they are all equally good.

2.8 The moduli problem

What happens to the physics if there are all these scalars in four dimensions? The answer is that the moduli mediate new interactions. Since in ten dimensions they are the metric, the moduli couple gravitationally to everything. This also explains why we have not seen them: they carry no charge under the Standard Model gauge group, and feel neither QCD nor the electroweak interactions. But the gravitational interaction between two massive particles is the exchange of a graviton, and on top of it there can be the exchange of any of these scalars: a fermion can emit a modulus through a gravitational vertex, and another fermion can absorb it. This is a new interaction of gravitational strength, which modifies Newton's law.

How far can this be tolerated? Newton's inverse-square law $1/r^2$ has been verified experimentally down to distances of a few tens of micrometres, which is not small at all: below that we do not know how gravity behaves, whereas the LHC probes the other interactions down to 10^{-18} m. The reason is that when two bodies are pushed closer to each other the other interactions become more important, and they are hard to shield. What this means depends on the masses of the moduli. Massless moduli modify gravity at arbitrarily large distances. Massive ones mediate a force of finite range, just as the weak interactions are felt only inside nuclei because the W and Z bosons are massive, and modify gravity only at short distances. To avoid these *fifth forces* a string compactification must give all the moduli a mass large enough for the forces to have escaped detection. At tree level, however, all of them are massless, with nothing to give them a mass, so one has to include quantum corrections and see whether they generate a potential that does. This is the problem of *moduli stabilisation*. A massless field has a flat potential, so its value is not fixed anywhere and can have very large quantum fluctuations; a potential with a minimum stabilises the field there, besides giving it a mass.

In summary, *moduli are uncharged four-dimensional scalars with gravitational-strength interactions with ordinary matter, since they come from the metric, and are massless at leading (tree-level) order in supergravity.* Their potential is flat: deformations of the size and shape of the extra dimensions cost no energy. This would be a phenomenological disaster, for two reasons.

2.8.1 Fifth forces

The first reason is the one just described: massless moduli would mediate fifth forces of Yukawa type, which are not observed. The four-dimensional Lagrangian contains, together with the mass term of a fermion ψ of ordinary matter, such as the electron, an interaction of Yukawa type with a modulus,

$$\mathcal{L}_4 \supset \frac{m_\psi}{M_P}\, \phi\, \bar{\psi}\psi, \tag{2.15}$$

where ϕ stands for any of the moduli, canonically normalised, and the coupling up to a coefficient of order one is shown. The suppression by the Planck mass is typical of gravitational interactions. Through this coupling the exchange of ϕ between two particles of ordinary matter, Figure 2.3, gives a fifth force, with a vertex m_ψ/M_P at each end. The

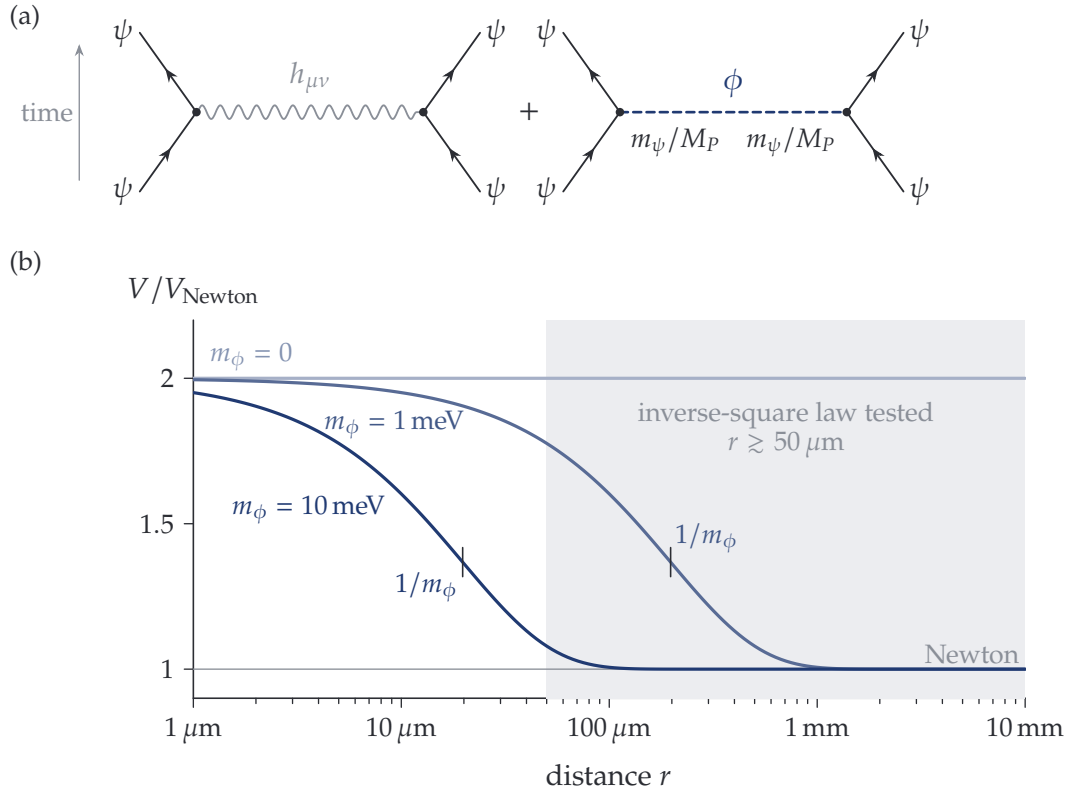


Figure 2.3: A modulus as the mediator of a fifth force. (a) Tree-level exchange of a modulus ϕ between two fermions, with the coupling of Equation (2.15) at each vertex, on top of the exchange of a graviton. (b) The resulting potential between two masses relative to Newton's, $V/V_{\text{Newton}} = 1 + \alpha_\phi\, e^{-m_\phi r}$, where α_ϕ is the strength of the fifth force relative to gravity (here $\alpha_\phi = 1$), for a massless modulus and for two massive ones.

observational limit for moduli that couple to ordinary matter with strength $1/M_P$ is that, to avoid these fifth forces, they must be heavier than about a meV:

$$m_\phi \gtrsim 1\,\text{meV}. \tag{2.16}$$

2.8.2 Lack of predictivity

The second reason is that all the features of the effective theory depend on the moduli. The only parameter that can be fixed by hand is α', and once it is fixed (or traded for M_P, Equation (1.11)) everything else is the vacuum expectation value of a field. For example, the coupling α_{GUT} of Equation (2.7), written in terms of the moduli with Equation (2.14), is

$$\alpha_{\text{GUT}} \simeq \frac{g_s^2}{\mathcal{V}} = \frac{1}{\langle\mathcal{V}\rangle\,\langle \text{Re}\, S\rangle^2}, \tag{2.17}$$

where $\langle\mathcal{V}\rangle$ is the expectation value of the volume modulus: the coupling is the expectation value of a combination of two moduli. If these are massless, their potential is flat, any value is allowed, with huge quantum fluctuations, and α_{GUT} cannot be predicted. The same holds for the scales of Chapter 1: the string scale $M_s \simeq g_s M_P/\sqrt{\mathcal{V}}$ (Equation (1.11)) and the Kaluza–Klein scale $M_{KK} \simeq g_s M_P/\mathcal{V}^{2/3}$ (Equation (1.14)) are other combinations of g_s and $\mathcal{V}$. Without knowing the moduli one knows neither M_s, nor M_{KK}, nor α_{GUT}: one knows nothing.

There is therefore a huge lack of predictivity, because all the features of the four-dimensional effective theory are functions of the moduli: the gauge couplings, the Yukawa couplings, the mass spectrum, the string and Kaluza–Klein scales, the scale of supersymmetry breaking, the cosmological constant, the scale of inflation, and even the gauge group. In a sense this is the beauty of string theory, since all these quantities can be predicted rather than chosen by hand; but with massless moduli nothing is predicted.

What is needed is *moduli stabilisation*: the study of the quantum corrections to the scalar potential V (not to be confused with the volume $\mathcal{V}$), which can develop a minimum, stabilise the moduli ϕ^I there, make them massive and fix the predictions,

$$\left.\frac{\partial V}{\partial \phi^I}\right|_{\langle\phi\rangle} = 0, \qquad m^2_{IJ} \propto \left.\frac{\partial^2 V}{\partial \phi^I \partial \phi^J}\right|_{\langle\phi\rangle} \text{ positive definite,} \tag{2.18}$$

with the mass matrix normalised by the kinetic terms of the moduli, and with masses large enough to satisfy Equation (2.16) and the much stronger cosmological bound Equation (3.4) of Section 3.5. Why this is hard in the heterotic string is the subject of Section 3.1; a hint is already in Section 2.6: there the quantum corrections are hard to control, and yet they are exactly what is needed to stabilise the moduli.

Chapter 3

Moduli stabilisation: the Dine–Seiberg problem and the cosmological moduli problem

Lecture of 12 November 2026

This lecture completes the general picture drawn in the previous two. Many statements are still made without derivation, deliberately: the aim is an overall view of how string compactifications connect with the real world, and of what can go wrong, before the material is written down in equations, starting in the next lecture with D-branes. The lecture has three parts. The first explains why moduli stabilisation, introduced at the end of Chapter 2, is particularly hard in the heterotic string, and why type IIB is better placed. The second discusses, in a toy model with a single modulus, the general difficulty of stabilising the moduli with quantum corrections while keeping the effective theory under control, the *Dine–Seiberg problem*, and the idea that resolves it. The third explains, with a simple cosmological estimate, why the moduli must be heavier than about 30 TeV: the *cosmological moduli problem*.

3.1 Why moduli stabilisation is hard in the heterotic string

The moduli of Section 2.7, the Kähler moduli T_i, the complex-structure moduli U_a and the axio-dilaton S, have a flat potential at tree level, which leads to fifth forces and to a complete lack of predictivity (Section 2.8). Moduli stabilisation is the study of the quantum corrections that generate a potential for them, lift the flat directions and make them massive; the minimum of the potential then fixes, or *stabilises*, the moduli at a point where predictions become possible.

In the heterotic string this programme is problematic, for three reasons. The first was met in Section 2.6: by Equation (2.9) the string coupling cannot be small, to suppress higher string loops, while the volume is large, to suppress higher-derivative corrections, so the effective theory is hard to trust. In type II theories one can have both; this was known from the beginning, but since type II theories seemed to give no non-abelian gauge bosons in four dimensions (Section 2.3), the advantage did not seem to matter.

3.1.1 Background fluxes

The second reason is that a major source of the potential for the moduli are *background fluxes*, generalisations of electromagnetic fluxes. By Gauss's law, the flux of the electric field through a sphere surrounding a charge measures the charge, and similarly for the magnetic flux of a

monopole; such fluxes carry energy. In four-dimensional Minkowski space a background electromagnetic field would pick a direction and break Lorentz invariance, but in the extra dimensions nothing forbids it: turning on fluxes there is like giving the field a vacuum expectation value in the extra dimensions, leaving the four-dimensional Lorentz symmetry untouched. The energy of a flux through a sphere depends on the size of the sphere, and if the sphere is in the extra dimensions its size is set by the size and shape of Y_6, i.e. by the moduli. A non-zero flux is therefore a source of energy that depends on the moduli: a potential for them.

The fluxes of string theory are those of the p-form fields of Table 2.1. The simplest is that of the Kalb–Ramond field B_{MN}, present in every closed superstring theory, whose field strength is the antisymmetrised derivative

$$H_3 = \mathrm{d}B_2, \qquad H_{MNP} = 3\,\partial_{[M}B_{NP]} = \partial_M B_{NP} + \partial_N B_{PM} + \partial_P B_{MN}, \tag{3.1}$$

with forms identified with their components as in the conventions. This generalises the electromagnetic field strength, $F_2 = \mathrm{d}A_1$, i.e. $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$. The flux of H_3 goes not through a sphere but through a *three-cycle* Σ_3, a three-dimensional submanifold of Y_6 without boundary, something like a three-sphere. In the vacuum considered so far the flux was zero; the idea is now to allow

$$\int_{\Sigma_3} H_3 \neq 0. \tag{3.2}$$

The energy of such a flux depends on the volume of the three-cycle, and the volumes of three-cycles are controlled, for example, by the complex-structure moduli, which set the shape of the extra dimensions. Fluxes therefore generate a potential for the moduli, as needed.

The fluxes, however, are sources of energy in the extra dimensions, and energy curves spacetime: the fluxes *backreact* on the Calabi–Yau. A mild possibility is *warping*, in which the metric changes only by a conformal factor and the internal space stays conformally Calabi–Yau. In the heterotic string this is not what happens in general: the backreaction is so strong that the extra dimensions are no longer a Calabi–Yau manifold, and the chain of consequences is

non-zero fluxes backreact on Y_6
$\Downarrow$
unknown moduli space
$\Downarrow$
loss of control of the effective theory.

Without control of the geometry one cannot even write down the tree-level effective theory. These manifolds are generalisations of Calabi–Yau manifolds that mathematicians are still trying to understand. Such compactifications are not wrong, but at present it is hard to compute with them. In type IIB (but not in type IIA) this problem is absent: fluxes can be turned on to stabilise moduli without destroying the Calabi–Yau background.

3.1.2 The visible sector and the moduli in the same bulk

The third reason is that in the heterotic string all the degrees of freedom, those of the Standard Model as well as the moduli, come from closed strings and live in the bulk of Y_6. In type IIB the moduli are again closed-string modes, but the Standard Model cannot be, since the closed-string spectrum has no non-abelian gauge bosons (Section 2.4); it comes instead from open strings on D-branes (Sections 2.3 and 4.2). The two tasks, reproducing the Standard Model and stabilising the moduli, then concern two different kinds of strings,

which decouple at leading order: one can solve each problem separately and then combine the solutions into a consistent whole. In the heterotic string both have to be solved at once, which is clearly harder. It is hard to address at the same time

1. the construction of the Standard Model, with the correct gauge group, the chiral spectrum and the Yukawa couplings, and
2. moduli stabilisation.

The heterotic string remains an active field; the point here was how these problems arose historically. In type IIB the first problem is absent, the fluxes do not destroy the Calabi–Yau, and the two tasks of the third decouple. Before turning to type IIB compactifications, we discuss the problems of moduli stabilisation in general, from the point of view of the effective theory.

3.2 What moduli stabilisation requires

In general there are hundreds of moduli, and models with many of them are very hard to discuss. A very simplified toy model already shows the main issues. Let ϕ denote collectively all the moduli; for simplicity we mostly consider a single one. (In this chapter ϕ is a generic modulus, not the dilaton; see the conventions.) At tree level the potential vanishes,

$$V_{\text{tree}}(\phi) = 0. \tag{3.3}$$

The quantum corrections must make it non-zero, $V(\phi^I) \neq 0$, and develop a minimum $\langle\phi^I\rangle$ with no flat direction, as in Equation (2.18), so that all the moduli are massive, $m^2_{\phi^I} > 0$. The masses must also be large enough: above about a meV, Equation (2.16), to avoid fifth forces, and, by a much stronger bound derived in Section 3.5,

$$m_{\phi^I} \gtrsim 30\,\text{TeV} \qquad \text{for all } \phi^I, \tag{3.4}$$

to avoid the so-called *cosmological moduli problem*.

The bound can be evaded or weakened in some models, but its origin is generic. Scalar fields with gravitational couplings easily come to dominate the energy density of the early universe: they behave as matter, so that between inflation and the late epochs of matter and cosmological-constant domination they produce an extra epoch dominated by the moduli. If stable, they overclose the universe; if they decay while dominating, the radiation they release dilutes everything produced before, and spoils Big Bang nucleosynthesis, whose predictions for the abundances of the light elements fit the data extremely well. Moduli heavier than about 30 TeV decay before nucleosynthesis starts, and are harmless for it. The argument is made quantitative in Section 3.5.

This is a very large mass for fields that are massless at tree level. It also has an interesting connection with supersymmetry: the effects that break supersymmetry generally also give mass to the moduli, and if the same mechanism gives mass to the squarks, the sleptons and the gauginos, the cosmological bound may indicate that the supersymmetric particles are heavy too, at least some 30 to 100 TeV.

Finally, the minimum must lie where the effective theory is under control. Since the strength of the couplings is itself set by the moduli, one proceeds as in Section 1.4: assume weak coupling, keep only the leading quantum corrections, compute the potential they generate, and check a posteriori that its minimum is at weak coupling; if it is not, the result must be thrown away. The difficulty of meeting this condition is the *Dine–Seiberg problem* [11], made explicit by the toy model of Section 3.4. Part of the difficulty is that $V_{\text{quantum}}(\phi)$, unlike $V_{\text{tree}} = 0$, requires computations beyond tree level.

3.3 Three kinds of quantum corrections

In an effective theory derived from string theory the potential receives, beyond tree level, three types of quantum corrections:

$$V_{\text{quantum}} = V_{g_s} + V_{\alpha'} + V_{\text{np}}. \tag{3.5}$$

The first are loop corrections, as in quantum field theory, controlled by the string coupling. The second are the higher-derivative α' corrections of Section 1.5, present already at tree level in g_s, which vanish in the point-particle limit $\alpha' \to 0$. Both are perturbative: the first are loops in spacetime, the second loops in the worldsheet theory. The third are non-perturbative, both in g_s and in α'.

String loops. The string-loop corrections scale as

$$V_{g_s} = \sum_{m\geq 1} g_s^m \, V_{(m)}(\text{moduli}), \tag{3.6}$$

with m the number of loops and coefficients $V_{(m)}$ that depend on the moduli. But g_s is itself a modulus, $g_s = 1/\langle \text{Re}\, S\rangle$ by Equation (2.14): here the field ϕ of the toy model is $\text{Re}\, S$, and to trust the expansion one needs

$$g_s \ll 1 \qquad \Longleftrightarrow \qquad \text{Re}\, S \gg 1. \tag{3.7}$$

All the loops cannot be computed; in practice only the one-loop term can, and one works with it alone, assuming the higher ones to be small. The procedure is consistent if the resulting potential has a minimum at $\text{Re}\, S \gg 1$; if instead it just runs away in that region, two loops are needed as well, and so on.

In terms of $\phi = \text{Re}\, S$, the loop expansion Equation (3.6) becomes a series in powers of $1/\phi$,

$$V_{g_s} = \frac{c_1}{\phi} + \frac{c_2}{\phi^2} + \frac{c_3}{\phi^3} + \dots, \tag{3.8}$$

whose terms go like g_s, g_s^2, g_s^3, and so on, and the problem is immediately visible. The hard part is to compute the coefficients. If only c_1 is known, the potential c_1/ϕ has no minimum: it is a runaway. A minimum needs the second term as well, and if it comes from balancing the first term against the second, then at the minimum one loop and two loops are of the same size. For the minimum to be at $\phi \gg 1$, c_2 must even be much bigger than c_1 in absolute value (and of opposite sign, for the two terms to balance). But if two loops are as big as one loop, why should three loops be negligible?

α' corrections. The higher-derivative corrections have a similar structure,

$$V_{\alpha'} = \sum_{m\geq 2} \alpha'^m \, V_{(m)}, \tag{3.9}$$

but the sum now starts from $m = 2$, because supersymmetry makes the leading corrections vanish. Since α' is dimensionful, it must be compared with the size of the extra dimensions, and by Equation (1.13)

$$\alpha' \ll \text{Vol}(Y_6)^{1/3} \qquad \Longleftrightarrow \qquad \mathcal{V} \gg 1 \tag{3.10}$$

parametrically: as in Section 1.5, the α' expansion is under control when the volume is large in string units. The volume is a function of the Kähler moduli, so here the field ϕ of the toy model is $\mathcal{V}$: each power of α' comes with a power of $1/\mathcal{V}^{1/3}$, and $V_{\alpha'}$ is again an expansion in inverse powers of a modulus, like Equation (3.8). The two expansions are intertwined:

there is a whole α' expansion at tree level in g_s, another at one loop, and so on, and a generic correction is of a given order in both.

Non-perturbative corrections. The last piece are corrections non-perturbative in g_s or in α', the latter called *worldsheet instantons*. Those non-perturbative in g_s also form an expansion,

$$V_{\text{np}} = \sum_{k\geq 1} e^{-kA_k/g_s}\, V_{(k)}, \tag{3.11}$$

with positive constants A_k and coefficients $V_{(k)}$ that can depend on the moduli. This expansion, too, can be trusted at small g_s, where the exponentials are very small. If moreover the A_k are bigger than one, each further term carries an additional exponential suppression, e^{-A/g_s}, then e^{-2A/g_s}, and so on, and one can keep the leading term.

3.4 The Dine–Seiberg problem

To illustrate the Dine–Seiberg problem, consider the perturbative expansion alone, as one would naively do, since perturbative corrections are expected to dominate over non-perturbative ones, which are negligible if the constants A_k in Equation (3.11) are big. The potential then has the form

$$V(\phi) = \frac{a}{\phi} - \frac{b}{\phi^2} + \frac{c}{\phi^3} + O\Big(\frac{1}{\phi^4}\Big), \tag{3.12}$$

where ϕ can be either $\operatorname{Re} S$ or the volume $\mathcal{V}$, both dimensionless, and the minus sign of the second term is written explicitly, so that the interesting case has $a, b, c > 0$. One expects the three coefficients to be of order one,

$$a \sim b \sim c \sim O(1), \tag{3.13}$$

as coefficients usually are. There are then two cases.

Trusting the effective theory: a runaway. The effective theory can be trusted for $\phi \gg 1$, a very big volume or a very small string coupling. But if all the coefficients are comparable, for $\phi \gg 1$ the first term dominates,

$$V(\phi) \simeq \frac{a}{\phi}, \tag{3.14}$$

a runaway, shown in Figure 3.1(a): the potential has no minimum, and the field is not stabilised. This need not be thrown away at once. At $\phi \gg 1$, in a region of positive vacuum energy, the field could roll very slowly down a shallow, almost flat potential, and explain the present acceleration of the universe: this is the idea of *quintessence*. But these corrections in general never give a potential flat enough for slow roll, and even if they did, the mass of the field would be of the order of the present Hubble scale, $H_0 \sim 10^{-33}$ eV, and the field would mediate the fifth forces of Section 2.8. So this does not really work.

Finding a minimum: tuning. The other possibility is to use all the terms to find a minimum. Setting the first derivative to zero,

$$V'(\phi) = -\frac{a}{\phi^2} + \frac{2b}{\phi^3} - \frac{3c}{\phi^4} = 0, \tag{3.15}$$

gives a quadratic equation, $a\phi^2 - 2b\phi + 3c = 0$, with the two solutions

$$\phi_{0,\pm} = \frac{b}{a}\Big(1 \pm \sqrt{1 - \frac{3ac}{b^2}}\Big), \tag{3.16}$$

which are real for $b^2 \geq 3ac$. To check which is a minimum and which a maximum, one computes the second derivative,

$$V''(\phi) = \frac{2a}{\phi^3} - \frac{6b}{\phi^4} + \frac{12c}{\phi^5}. \tag{3.17}$$

At a critical point, eliminating c with Equation (3.15), $V'' = 2(b - a\phi)/\phi^4$, and since $\phi_{0,-} < b/a < \phi_{0,+}$, for $b^2 > 3ac$

$$V''(\phi_{0,-}) > 0 \quad \text{(minimum)}, \qquad V''(\phi_{0,+}) < 0 \quad \text{(maximum)}. \tag{3.18}$$

To reproduce the cosmological constant, which today is almost zero, one also imposes that the potential vanish at $\phi_{0,-}$:

$$V(\phi_{0,-}) = 0 \qquad \Longleftrightarrow \qquad ac = \frac{b^2}{4}. \tag{3.19}$$

(Indeed, $V = V' = 0$ requires $a\phi^2 - b\phi + c = 0$ together with $a\phi^2 - 2b\phi + 3c = 0$, whose difference gives $\phi = 2c/b$, and substituting back gives $4ac = b^2$.) The potential then looks as in Figure 3.1(b), and seems to reproduce the real world: the field is stabilised at the minimum, the vacuum energy there vanishes, and the potential is steep around it, so that the fluctuations have a large mass and can satisfy the bounds of Section 3.2.

Can one then be in the limit $\phi \gg 1$? With Equation (3.19), the expression Equation (3.16) for the minimum becomes

$$\phi_{0,-} = \frac{b}{2a} \gg 1 \qquad \text{if} \qquad b \gg a, \qquad \text{and then} \qquad c = \frac{b^2}{4a} \gg b \gg a. \tag{3.20}$$

This contradicts Equation (3.13): one needs a hierarchy among the coefficients, $c \gg b \gg a$, which is a tuning. Is this natural? Is it even feasible? Moreover, the three contributions to the potential at the minimum are

$$V(\phi_{0,-}) = \frac{a}{\phi_{0,-}} - \frac{b}{\phi_{0,-}^2} + \frac{c}{\phi_{0,-}^3} = \frac{2a^2}{b} - \frac{4a^2}{b} + \frac{2a^2}{b} = 0. \tag{3.21}$$

They cancel, but they are all of the same order, a^2/b. Why, then, should the term in $1/\phi^4$ be negligible? The only reason why one could still be at weak coupling, ignoring four loops, five loops and so on, is the tuning that makes ϕ big, $c \gg b \gg a$; but nothing guarantees that the same tuning does not also enhance the coefficient of $1/\phi^4$. Either one can trust the effective theory, and then there is no minimum, or one finds minima by balancing different terms of the same expansion, and then the result is always dirty. This is the *Dine–Seiberg problem*, named after the paper that pointed it out [11]. It is not specific to the heterotic string, but a general problem of moduli stabilisation.

3.4.1 The way out: balancing different expansions

The Dine–Seiberg problem has been solved, by constructions found from the early 2000s onwards, which will be studied in equations towards the end of the course. The way out uses the fact that in string theory there is not just one expansion of the quantum corrections but three, the g_s, α' and non-perturbative expansions of Equation (3.5). Each of them can be under control, with its first term much bigger than its second, and yet the first term of one expansion can balance the first term of another. A minimum found in this way can be fully under control. The idea is therefore to balance not different terms of the same expansion, but

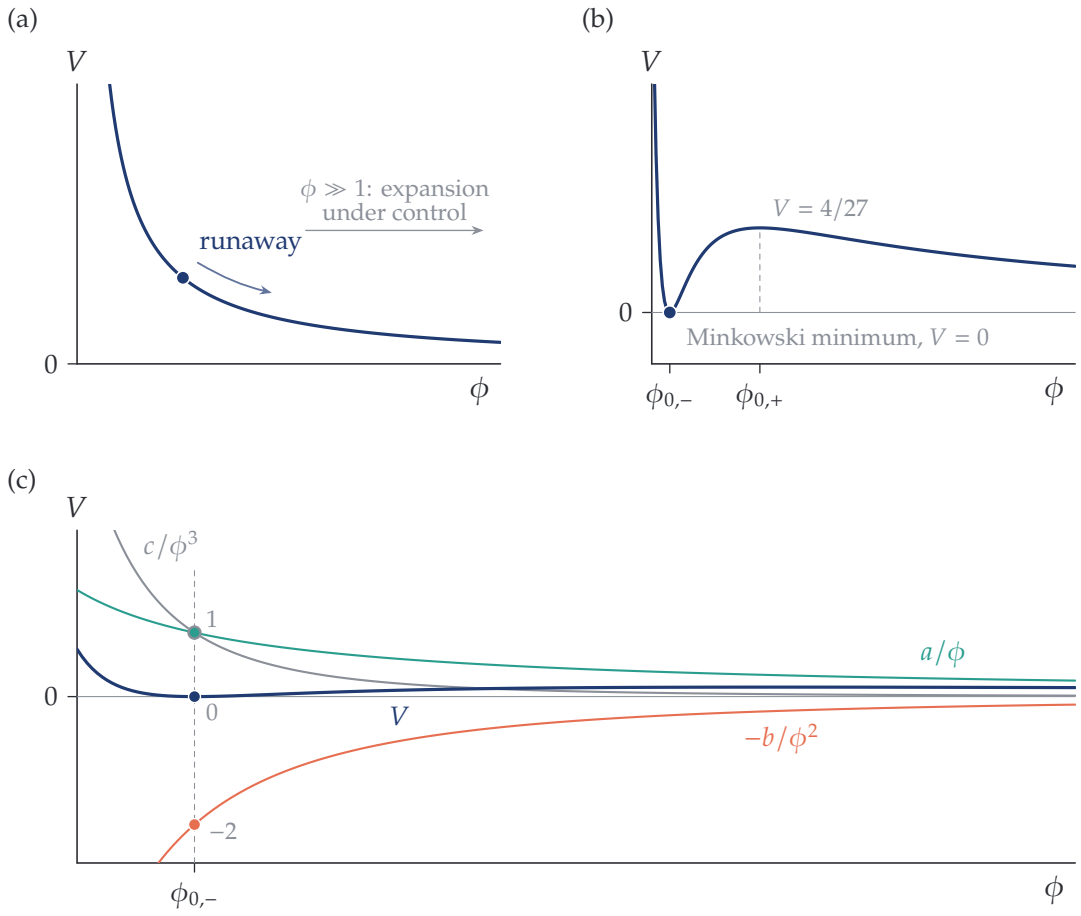


Figure 3.1: The Dine–Seiberg problem in the toy model Equation (3.12). (a) Where the expansion can be trusted, $\phi \gg 1$, only the leading term survives, $V \simeq a/\phi$, and the modulus runs away to $\phi \to \infty$. (b) Keeping three terms, with $ac = b^2/4$, gives a Minkowski minimum at $\phi_{0,-} = b/2a$ and a maximum at $\phi_{0,+} = 3b/2a$. (c) At the minimum the three terms are of the same size, here 1, −2 and 1, and cancel.

terms of different expansions: for example α' corrections against string-loop corrections, or perturbative against non-perturbative corrections. There are two famous solutions of this kind. One is *KKLT*, from the initials of its authors, Kachru, Kallosh, Linde and Trivedi [12]. The other, which appeared shortly afterwards, is the *Large Volume Scenario* (LVS) [13], in which, as the name says, the volume of the extra dimensions is very large.

3.5 The cosmological moduli problem

We now derive the bound Equation (3.4). There is no string theory in the argument, only the cosmology of a heavy scalar field with gravitational couplings, which must decay before Big Bang nucleosynthesis.

3.5.1 A modulus displaced by inflation

Suppose that the quantum corrections, in a scenario such as KKLT or the LVS, have produced a minimum in a regime in which the effective theory can be trusted, and shift the field so that the minimum is at zero. However complicated the potential is, around the minimum it is quadratic,

$$V(\phi) = \frac{1}{2} m_\phi^2 \phi^2 \qquad \text{(at zero temperature)}, \tag{3.22}$$

where ϕ is now the canonically normalised modulus, of mass dimension one, and m_ϕ the mass of this spin-zero particle, which interacts gravitationally with everything.

In the very early universe, during inflation, the inflationary energy density, set by the Hubble rate as $\rho_{\text{inf}} = 3H^2 M_P^2$, contributes to the potential for the moduli, as any source of energy does, since the moduli couple gravitationally to it. A toy model of its effect is

$$V(\phi) = \frac{1}{2} m_\phi^2 \phi^2 + \frac{1}{2} H^2 (\phi - \phi_0)^2 \simeq \frac{1}{2} H^2 (\phi - \phi_0)^2 \qquad \text{for } H \gg m_\phi, \tag{3.23}$$

where the second term is the effect of inflation, with a coefficient of order one in general, here set to one. During inflation H is much bigger than m_ϕ: the inflationary Hubble scale can be as high as 10^{13}–10^{14} GeV (present bounds on primordial gravitational waves require $H \lesssim 5 \times 10^{13}$ GeV), while the modulus is much lighter. The minimum is then shifted to $\phi = H^2 \phi_0 / (H^2 + m_\phi^2) \simeq \phi_0$, as in Figure 3.2: in the early universe the field sits not at zero but at ϕ_0. Since everything here is gravitational, the shift is expected to be of the order of the Planck scale, $\phi_0 \sim M_P$.

The evolution of the field is governed by the Klein–Gordon equation in an expanding universe, with the Friedmann–Lemaître–Robertson–Walker metric of the conventions. For a field that is homogeneous in space it reads

$$\ddot{\phi} + 3H\dot{\phi} + m_\phi^2 \phi = 0. \tag{3.24}$$

For $H = 0$ this is a harmonic oscillator. The term $3H\dot{\phi}$, due to the expansion of the universe, acts as friction, and for $H \gg m_\phi$ it dominates: the field is stuck at ϕ_0, high up in the potential. As the universe expands, H decreases, the potential approaches the zero-temperature one, Equation (3.22), and the friction becomes less and less important compared with the restoring force, until H becomes of order m_ϕ and the field starts oscillating around the minimum:

$$H_{\text{osc}} \simeq m_\phi. \tag{3.25}$$

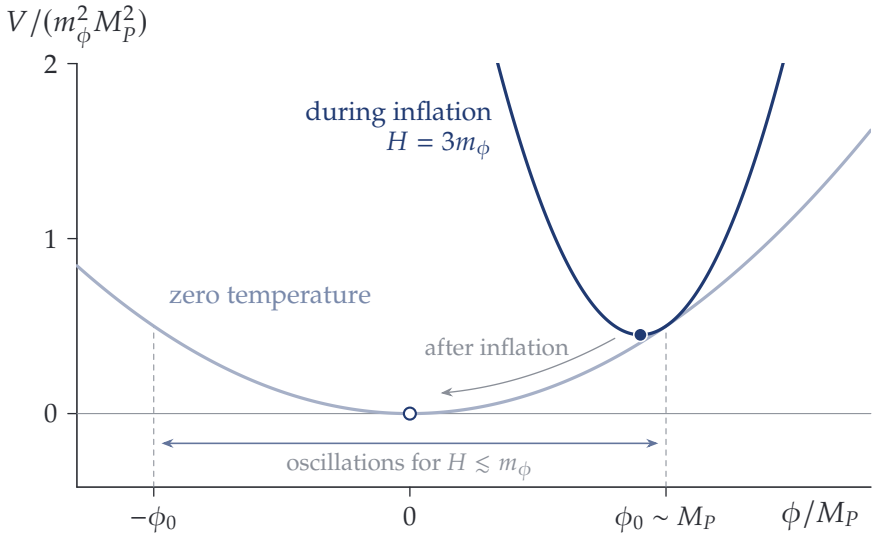


Figure 3.2: A modulus during and after inflation. At zero temperature its potential is $\frac{1}{2}m_\phi^2\phi^2$, with the minimum at $\phi = 0$ (light curve); during inflation the Hubble-induced term of Equation (3.23), with $H \gg m_\phi$ (here $H = 3m_\phi$), moves the minimum to $\phi \simeq \phi_0$ (dark curve). After inflation the field is left at ϕ_0, far from the late-time minimum, and starts oscillating around it when $H \simeq m_\phi$.

The initial amplitude is ϕ_0, so the oscillations are large and the field stores energy instead of sitting at the minimum. While the field is stuck, its energy is only a small contribution to the vacuum energy; at the onset of the oscillations it is of the order of the potential,

$$\rho_\phi^{\text{osc}} \simeq m_\phi^2\phi_0^2 \simeq m_\phi^2 M_P^2. \tag{3.26}$$

3.5.2 Modulus domination and decay

This energy density has to be compared with that of the standard cosmological history. After inflation the inflaton decays and reheats the universe: its energy goes into the particles of the Standard Model, which form a thermal bath and redshift as radiation. While the modulus is stuck it carries no energy density of its own, and all the energy is in radiation. But when H drops to m_ϕ, during radiation domination, part of the energy is in the oscillating ϕ. The energy density of radiation at that time follows from the Friedmann equation, $3H^2M_P^2 = \rho$:

$$\rho_\gamma^{\text{osc}} \simeq H_{\text{osc}}^2 M_P^2 \simeq m_\phi^2 M_P^2. \tag{3.27}$$

The two energy densities, Equations (3.26) and (3.27), are comparable: the modulus, which had no energy of its own, has as much energy as the whole Standard Model as soon as it starts oscillating.

As the universe expands both energy densities are diluted, but in different ways. A field oscillating coherently around its minimum is a condensate, a large number of quanta with zero momentum, and so it behaves as matter, which redshifts more slowly than radiation,

$$\rho_\phi \propto a^{-3}, \qquad \rho_\gamma \propto a^{-4}, \tag{3.28}$$

where the scale factor a grows with the expansion. Since the two start with comparable energies, ϕ very quickly comes to dominate the energy density of the universe.

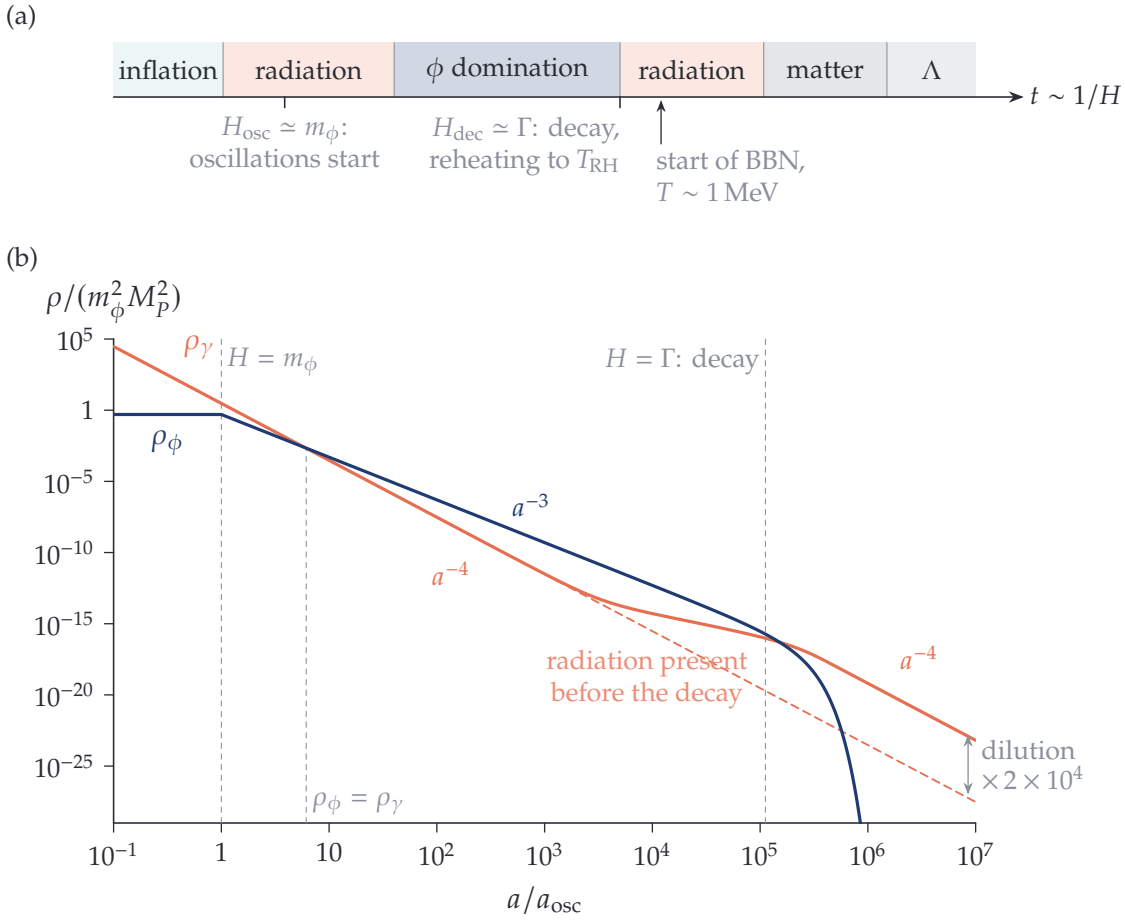
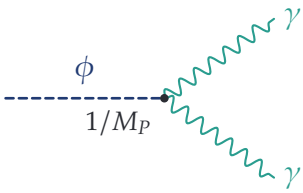


Figure 3.3: Cosmological history with a displaced modulus. (a) The sequence of epochs: inflation, radiation domination, domination by the oscillating modulus, radiation domination again after its decay, then matter and cosmological-constant domination. The modulus must decay before Big Bang nucleosynthesis starts. (b) Energy densities of the modulus and of radiation as functions of the scale factor, from the onset of the oscillations at $H_{\rm osc} \simeq m_\phi$ to the decay at $H_{\rm dec} \simeq \Gamma$; the dashed line is the radiation present before the decay, diluted by it. The ratio $\Gamma/m_\phi = 10^{-8}$ is chosen for illustration; for a real modulus it is $(m_\phi/M_P)^2 \sim 10^{-28}$.

This is the problem anticipated in Section 3.2. A stable modulus carries so much energy that it overcloses the universe, as too much dark matter. A modulus that decays into Standard Model particles converts all its energy into radiation, a huge injection of energy that dilutes all the particles present before, and with them the successful predictions for the abundances of the light elements. The decay happens when H becomes of order the decay rate of ϕ,

$$H_{\rm dec} \simeq \Gamma_{\phi \to \rm SM}. \tag{3.29}$$

since $1/H$ is the age of the universe and $1/\Gamma = \tau$ the lifetime of the particle. The history of the universe is then modified as in Figure 3.3(a): after inflation and radiation domination comes an epoch of ϕ domination; when ϕ decays the universe returns to radiation domination, and then to matter and cosmological-constant domination as usual. Since everything produced before the decay is diluted, the decay must happen before Big Bang nucleosynthesis starts.



$$\Gamma \simeq \frac{m_\phi^3}{M_P^2}$$

Figure 3.4: Decay of a modulus into two gauge bosons through the coupling Equation (3.30), with one vertex suppressed by $1/M_P$, which gives $\Gamma \simeq m_\phi^3/M_P^2$ (Equation (3.31)).

3.5.3 The bound on the mass of the moduli

To estimate $\Gamma_{\phi\to\text{SM}}$, note that the moduli couple to the gauge bosons: the gauge kinetic function, a function of the fields as seen in the supersymmetry course, depends in string compactifications on the moduli. Expanding it around the minimum gives a coupling of ϕ to two gauge bosons, suppressed by the Planck mass,

$$\mathcal{L}_4 \supset \frac{\phi}{M_P} F_{\mu\nu}F^{\mu\nu}, \tag{3.30}$$

up to a coefficient of order one, through which the modulus decays as in Figure 3.4, very weakly. Dimensional analysis is enough to estimate the rate. It goes like the square of the coupling, $1/M_P^2$, and has dimensions of a mass, so it must contain a mass cubed, and the only mass available in the decay into massless gauge bosons is m_ϕ:

$$\Gamma_{\phi\to\text{SM}} \simeq \frac{m_\phi^3}{M_P^2}. \tag{3.31}$$

The Hubble rate at the decay is then

$$H_\text{dec} \simeq \Gamma_{\phi\to\text{SM}} \simeq m_\phi \left(\frac{m_\phi}{M_P}\right)^2 \simeq H_\text{osc}\left(\frac{m_\phi}{M_P}\right)^2 \ll H_\text{osc}, \tag{3.32}$$

by Equation (3.25), since the moduli are much lighter than the Planck scale: the moduli decay long after they start oscillating.

The decay reheats the universe, and H_dec is the Hubble rate at the start of the new epoch of radiation domination, $H \simeq T^2/M_P$ (see the conventions), at a temperature called the *reheating temperature* T_RH. Hence

$$H_\text{dec} \simeq \Gamma_{\phi\to\text{SM}} \simeq \frac{T_\text{RH}^2}{M_P} \qquad \Longrightarrow \qquad T_\text{RH} \simeq \sqrt{\Gamma_{\phi\to\text{SM}} M_P} \simeq m_\phi \sqrt{\frac{m_\phi}{M_P}}, \tag{3.33}$$

where the last step uses Equation (3.31).

For the decay to happen before Big Bang nucleosynthesis, the reheating temperature must exceed the temperature at which the light elements started to form, of order one MeV:

$$T_\text{RH} > T_\text{BBN} \sim 1\,\text{MeV} \sim 10^{-21} M_P. \tag{3.34}$$

With Equation (3.33) this gives $m_\phi^{3/2} \gtrsim 10^{-21} M_P^{3/2}$, i.e.

$$m_\phi \gtrsim 10^{-14} M_P \sim 10^4\,\text{GeV}, \tag{3.35}$$

a mass of order 10 TeV. A more precise computation, keeping all the factors dropped in the signs $\simeq$, gives 30 TeV: this is where the bound Equation (3.4) comes from.

String compactifications must therefore make the moduli quite heavy. The bound can be evaded if the cosmological history has new features. The estimate assumed an initial displacement ϕ_0 of order M_P; if some mechanism made it much smaller, the energy stored in ϕ and the dilution would be much smaller too, and the cosmological history would have to be studied in detail. Alternatively, the displacement could be of order M_P but the moduli themselves diluted by something else in some epoch. So the bound Equation (3.4) is much stronger than the bound Equation (2.16) from fifth forces, and generically present, but it depends somewhat on the cosmology, whereas the much weaker one from fifth forces rests on observations made today.

The next lecture, Chapter 4, starts looking at type IIB in equations with D-branes and their effective action; the bulk action, and how Calabi–Yau manifolds emerge, will follow.

Chapter 4

D-branes, differential forms and electric–magnetic duality

Lecture of 13 November 2026

With this lecture we start writing type IIB string theory in equations. The first ingredient is the D-brane: what it is, why it made type II theories the natural arena for string phenomenology, and the effective action that describes it. That action is written in the language of differential forms, which are introduced next, operatively, and used to rewrite Maxwell's theory. In that language electric–magnetic duality becomes manifest, and with it the Dirac quantisation condition, which the next lecture generalises to branes.

4.1 Open strings and D-branes

As recalled in Section 2.3, type II compactifications became the main arena of string phenomenology after the discovery of D-branes around 1995: type II theories contain, besides closed strings, open strings whose endpoints lie on D-branes, and open strings can give fields in the adjoint representation of a non-abelian group, as gauge bosons are.

D-branes also supply a missing piece in the interpretation of the R–R forms of Table 2.1, which are present only in type IIA and type IIB. These forms generalise the electromagnetic potential: they are gauge fields. Gauge fields alone, however, are not the whole story: the electromagnetic field is sourced by charged particles, and the consistent theory, quantum electrodynamics, contains both. The heterotic string has no R–R forms and the question does not arise; in type II theories the objects that source the R–R forms are the D-branes.

4.1.1 Neumann and Dirichlet boundary conditions

Let $X^M(\sigma,\tau)$ be the position in spacetime of the point σ of an open string at worldsheet time τ, with $\sigma \in [0,\ell]$. What happens at the endpoints is fixed by boundary conditions, and there are two possibilities. The first are *Neumann* boundary conditions,

$$\partial_\sigma X^a(\sigma,\tau)\big|_{\sigma=0,\ell} = 0, \qquad a = 0,\dots,p, \tag{4.1}$$

imposed here on $p+1$ of the coordinates, time included. They state that no momentum flows out of the string through its endpoints, which move freely, and in fact at the speed of light. This follows from one of the Virasoro constraints, which come from the equation of motion of the worldsheet metric,

$$\partial_\tau X^M \partial_\tau X_M + \partial_\sigma X^M \partial_\sigma X_M = 0. \tag{4.2}$$

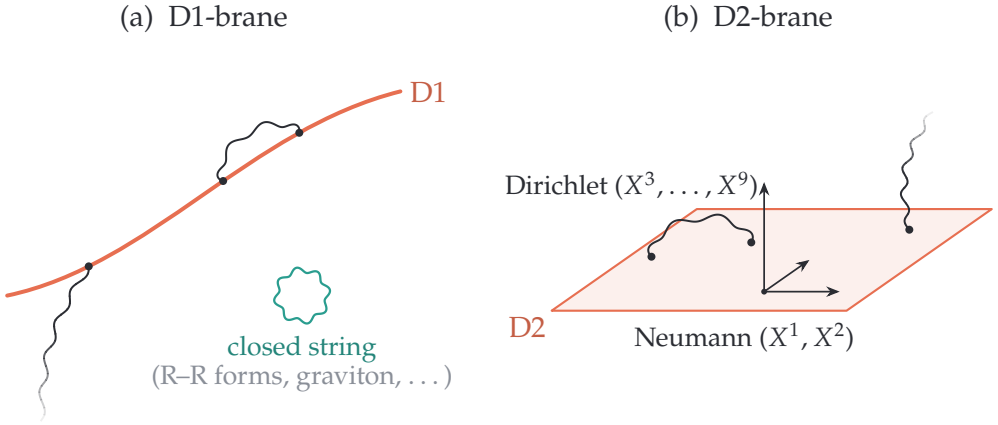


Figure 4.1: Open strings on D-branes. (a) A D1-brane, a string-like object, with an open string attached by both endpoints and one attached by a single endpoint; a closed string, such as a mode of the R–R forms, moves in the bulk. (b) A D2-brane, a plane in space: the open-string endpoints obey Neumann conditions along the plane and Dirichlet conditions in the seven transverse directions.

If all the coordinates obey Neumann conditions, the second term vanishes at the endpoints, so that there $\partial_\tau X^M \partial_\tau X_M = 0$. Choosing the worldsheet time to be the time of spacetime, $\tau = t$, and writing $X^M = (t, \vec{x})$, this is

$$\partial_\tau X^M \partial_\tau X_M = -1 + |\dot{\vec{x}}|^2 = 0 \qquad \Longrightarrow \qquad |\dot{\vec{x}}| = 1 \qquad \text{at } \sigma = 0, \ell, \tag{4.3}$$

so the endpoints move at the speed of light.

The second possibility are *Dirichlet* boundary conditions, which fix the position of the endpoints in the remaining directions,

$$X^i(\sigma, \tau)\big|_{\sigma=0,\ell} = X_0^i, \qquad i = p+1, \ldots, 9. \tag{4.4}$$

The endpoints are then fixed in $9-p$ space directions. Picture an open string whose endpoints must stay on the blackboard: they cannot leave it in the orthogonal direction, where they obey Dirichlet conditions, but move along it, at the speed of light, where they obey Neumann conditions. Mixed boundary conditions of this kind, Neumann in some directions and Dirichlet in the others, define a hyperplane, $X^i = X_0^i$, on which the open strings end.

This hyperplane is a *Dp-brane*, D for Dirichlet: a $(p+1)$-dimensional object, counting time. It is not a surface put in by hand but a dynamical object, which moves and fluctuates, and whose excitations are the open strings attached to it. We shall see in Section 4.3 that D-branes are non-perturbative states of the theory, which appear once non-perturbative physics is taken into account and can then be present in the background.

A D0-brane, with $p = 0$, is a particle, one-dimensional in the sense that its worldline is. A D1-brane is a string-like object, with a two-dimensional worldvolume, to which open strings can be attached, as in Figure 4.1(a). A D2-brane is a plane in space, with a three-dimensional worldvolume: the open strings attached to it have Neumann conditions in the two space directions along the plane, in which they move at the speed of light, and Dirichlet conditions in the remaining $10 - (2+1) = 7$ directions (Figure 4.1(b)).

4.1.2 D-branes as sources of the R–R forms

The analogy with electromagnetism now becomes precise. A D0-brane is a particle, and a charged particle is the source of an electromagnetic field, described by a potential A_μ with

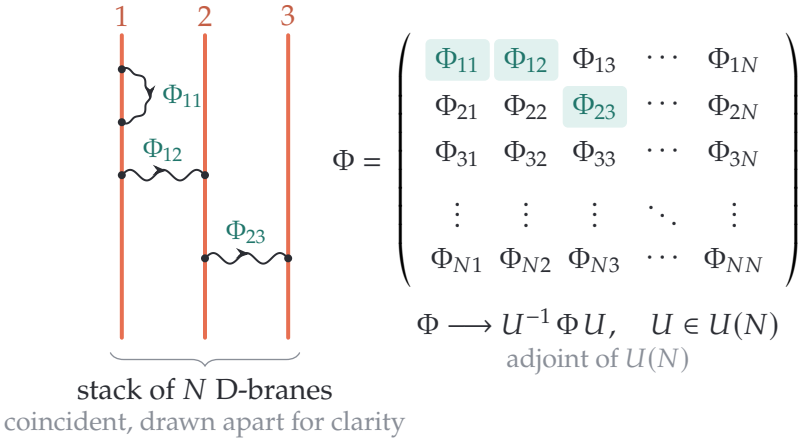


Figure 4.2: A stack of N coincident D-branes, drawn slightly apart. An open string from brane i to brane j gives the entry Φ_{ij} of the $N \times N$ matrix Equation (4.5), which transforms in the adjoint of $U(N)$.

one index and a field strength with two, $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$. An object extended in one direction is naturally the source of a potential with two indices, with a field strength with three, and so on: a Dp-brane, whose worldvolume has $p+1$ dimensions, sources a potential with $p+1$ indices, C_{p+1}, with field strength F_{p+2}. These potentials are the R–R forms of the massless spectrum, closed-string fields that propagate in the bulk of spacetime (Figure 4.1(a)).

4.2 Gauge theories on D-branes

The open strings, unlike the closed ones, live on the branes, and they give gauge theories. A single Dp-brane carries a $U(1)$ *vector multiplet*, in the sense of supersymmetry: among the massless modes of the open strings on it there is a $U(1)$ gauge field $A_a(\xi)$ that lives on the worldvolume, with field strength $F_{ab} = \partial_a A_b - \partial_b A_a$, together with the other fields of the multiplet.

A stack of N Dp-branes, all on top of each other at the same point of the transverse space, carries a vector multiplet in the adjoint representation of $U(N)$. To see why, label the branes $1, \ldots, N$, as in Figure 4.2. An open string can start on any brane i and end on any brane j: there are strings from 1 to 1, from 1 to 2, from 1 to 3, from 2 to 2, and so on, N^2 possibilities in all. The corresponding modes form an $N \times N$ matrix,

$$\Phi = \begin{pmatrix} \Phi_{11} & \Phi_{12} & \cdots & \Phi_{1N} \\ \Phi_{21} & \Phi_{22} & \cdots & \Phi_{2N} \\ \vdots & & \ddots & \vdots \\ \Phi_{N1} & \cdots & & \Phi_{NN} \end{pmatrix}, \tag{4.5}$$

whose labels i, j are called Chan–Paton indices. The branes of the stack are identical, so the labelling is arbitrary: it can be changed by a unitary transformation, under which

$$\Phi \longrightarrow \Phi' = U^{-1}\,\Phi\,U, \qquad U \in U(N), \tag{4.6}$$

which is how a field in the adjoint representation transforms. The massless vectors among these modes are the N^2 gauge bosons of $U(N)$.

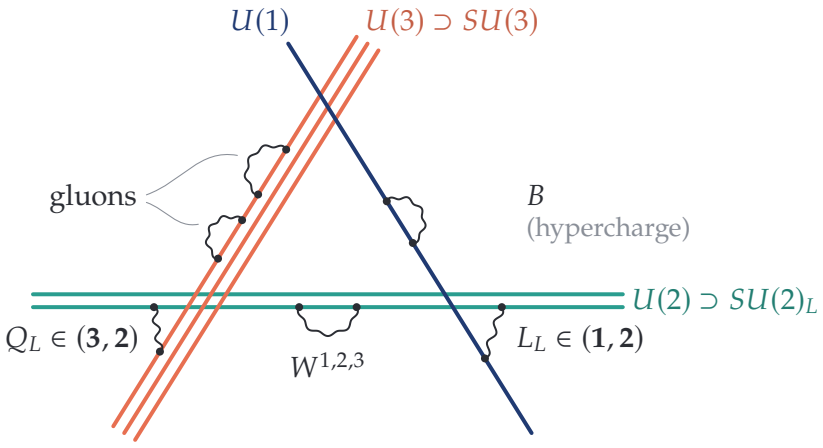


Figure 4.3: A cartoon of the Standard Model on intersecting D-branes: a stack of three branes ($U(3) \supset SU(3)$, gluons), a stack of two ($U(2) \supset SU(2)_L$, weak bosons) and a single brane ($U(1)$, hypercharge). Strings stretched between two stacks give fields in bifundamental representations: the quark doublet Q_L in $(\mathbf{3}, \mathbf{2})$ and the lepton doublet L_L.

4.2.1 The Standard Model on intersecting branes

The adjoint of $U(N)$ is exactly what a Yang–Mills theory needs, so D-branes can reproduce the gauge interactions of the Standard Model with Yang–Mills theories on their worldvolumes, as sketched in Figure 4.3. Take a stack of three branes, whose $U(3)$ contains the colour group $SU(3)$; a stack of two, whose $U(2)$ contains the weak $SU(2)_L$; and a single brane, with its $U(1)$, which plays the role of hypercharge. Open strings with both endpoints on the first stack are the gluons, those on the second stack the $SU(2)_L$ gauge bosons, and those on the single brane the hypercharge gauge boson. The $W^{\pm}$, the Z and the photon then arise, as usual, from $SU(2)_L \times U(1)_Y$ after electroweak symmetry breaking.

An open string can also stretch from one stack to another. A string from the $SU(3)$ stack to the $SU(2)$ stack is charged under both groups, in the bifundamental representation $(\mathbf{3}, \mathbf{2})$: it feels the strong interactions, so it is a quark, and it is a doublet of $SU(2)_L$, so it is left-handed. It can be the left-handed quark doublet Q_L. A string from the $SU(2)$ stack to the single brane feels $SU(2)_L$ and hypercharge but not colour: it can be the left-handed lepton doublet L_L. This is only a sketch, but it shows how open strings on D-branes can give something that looks like the Standard Model.

4.3 The D-brane effective action

The effective theory contains two kinds of fields. The closed strings move everywhere in spacetime, and their dynamics is described by the bulk action, studied later; the open strings are stuck on the branes, and their dynamics is described by an action localised on the brane worldvolume, with three pieces,

$$S_{\text{loc}} = S_{\text{DBI}} + S_{\text{CS}} + S_{\text{ferm}}. \tag{4.7}$$

The Dirac–Born–Infeld (DBI) action S_{DBI} describes the mass of the brane and the kinetic terms of the fields on it; the Chern–Simons (CS) action S_{CS} describes its coupling to the R–R forms. The theory is supersymmetric, so the action also contains the fermions, in S_{ferm}; we keep only the bosonic part, for a single brane rather than a stack.

4.3.1 The Dirac–Born–Infeld action

The DBI action is a generalisation of the Nambu–Goto action. The Nambu–Goto action of a string is proportional to the area of its worldsheet, and the action of a relativistic particle to the length of its worldline; the DBI action is, in the simplest case, proportional to the volume of the worldvolume of the brane. In general it reads

$$S_{\text{DBI}} = -T_p \int_{W_{p+1}} \mathrm{d}^{p+1}\xi \; e^{-\phi} \sqrt{-\det\bigl(g_{ab} + B_{ab} + 2\pi\alpha' F_{ab}\bigr)}. \tag{4.8}$$

Here W_{p+1} is the $(p+1)$-dimensional worldvolume of the Dp-brane, with coordinates ξ^a, $a = 0, \ldots, p$. It is embedded in spacetime: the point ξ of the worldvolume sits at the point $X^M = Y^M(\xi)$ of spacetime. The constant T_p is the *tension* of the brane,

$$T_p = \frac{1}{(2\pi)^p \, \alpha'^{(p+1)/2}} = 2\pi \, M_s^{p+1}, \tag{4.9}$$

where the second form follows from $M_s = 1/2\pi\sqrt{\alpha'}$ (Equation (1.4)). The dimensions are right: the action is dimensionless and the integral has dimensions of a length to the power $p+1$, so T_p must be a mass to the power $p+1$, and the only scale available is M_s. The field ϕ is the dilaton, and the action is written in the string frame.

The fields in the square root are of two kinds. $F_{ab} = \partial_a A_b - \partial_b A_a$ is the field strength of the $U(1)$ gauge boson that lives on the brane, an open-string field defined only on the worldvolume. The metric and the Kalb–Ramond field, instead, are closed-string fields, defined in the bulk of spacetime, and what enters Equation (4.8) are their *pullbacks* to the worldvolume:

$$g_{ab}(\xi) = G_{MN}\bigl(Y(\xi)\bigr) \, \partial_a Y^M \, \partial_b Y^N, \qquad B_{ab}(\xi) = B_{MN}\bigl(Y(\xi)\bigr) \, \partial_a Y^M \, \partial_b Y^N, \tag{4.10}$$

with $\partial_a = \partial/\partial\xi^a$: each field is evaluated where the brane is, which makes it a function of ξ, and its indices are converted into worldvolume indices with the derivatives of the embedding. For the metric this is the familiar induced metric, since the line element between two nearby points of the worldvolume is

$$\mathrm{d}s^2 = G_{MN} \, \mathrm{d}Y^M \mathrm{d}Y^N = \underbrace{G_{MN} \, \partial_a Y^M \partial_b Y^N}_{g_{ab}} \, \mathrm{d}\xi^a \mathrm{d}\xi^b. \tag{4.11}$$

4.3.2 D-branes are non-perturbative

In a vacuum the dilaton takes its expectation value, and the factor $e^{-\phi}$ in Equation (4.8) becomes $e^{-\phi_0} = 1/g_s$ (Equation (1.6)). The physical tension of the brane, its mass per unit volume, is therefore

$$T_p^{\text{phys}} = T_p \, e^{-\phi_0} = \frac{T_p}{g_s} = \frac{2\pi M_s^{p+1}}{g_s}. \tag{4.12}$$

In the limit of weak coupling, $g_s \to 0$, in which perturbation theory holds (Equation (1.7)), the tension diverges: the branes become infinitely massive and decouple. They become important as g_s grows, and light at strong coupling. D-branes are therefore non-perturbative objects, invisible in perturbation theory, which is why they were found only in the mid-1990s. Once non-perturbative effects are included, the theory turns out to contain them, and it needs them, since they are the sources of the R–R fields, as the Chern–Simons action shows.

4.3.3 The Chern–Simons action

The coupling of the brane to the R–R forms is

$$S_{\mathrm{CS}} = T_p \int_{W_{p+1}} C_{p+1}. \tag{4.13}$$

The same constant T_p appears again, and the case $p = 0$ shows what it means. For a particle, Equation (4.13) is the integral over the worldline of a one-form C_1, like the coupling $q \int A_\mu \, \mathrm{d}x^\mu = q \int A_1$ of a charged particle to the electromagnetic potential, whose coefficient is the charge. So T_p is not only the tension of the brane, which sets its mass, but also its charge under C_{p+1}. D-branes are particular states whose charge is equal to their mass, in appropriate units: they are *BPS states*.

For our purposes the R–R form C_{p+1} is just an abelian gauge potential with $p+1$ indices, with the gauge transformation

$$C_{p+1} \longrightarrow C'_{p+1} = C_{p+1} + \mathrm{d}\Lambda_p, \tag{4.14}$$

which generalises $A_\mu \to A_\mu + \partial_\mu \alpha$, i.e. $A_1 \to A_1 + \mathrm{d}\Lambda_0$, with $\Lambda_0 = \alpha$; the exterior derivative d is defined in Section 4.4. Since C_{p+1} is a closed-string field, living in the bulk, Equation (4.13) again involves its pullback to the worldvolume,

$$C_{a_1 \cdots a_{p+1}}(\xi) = C_{M_1 \cdots M_{p+1}}\big(Y(\xi)\big) \, \partial_{a_1} Y^{M_1} \cdots \partial_{a_{p+1}} Y^{M_{p+1}}. \tag{4.15}$$

The integral of a $(p+1)$-form over a $(p+1)$-dimensional worldvolume, and why it needs no measure $\mathrm{d}^{p+1}\xi$, is explained in the next section.

4.4 Differential forms

The Chern–Simons action is the integral of a differential form, and from now on the language of forms is used throughout. We introduce it operatively, without mathematical rigour.

A differential p-form is an antisymmetric tensor with p indices. On a D-dimensional manifold X with coordinates x^M it is written

$$C_p = \frac{1}{p!} \, C_{M_1 \cdots M_p}(x) \, \mathrm{d}x^{M_1} \wedge \cdots \wedge \mathrm{d}x^{M_p}, \tag{4.16}$$

where $\wedge$ is the *wedge product*, the antisymmetric part of the tensor product,

$$\mathrm{d}x^{M_1} \wedge \cdots \wedge \mathrm{d}x^{M_p} = p! \, \mathrm{d}x^{[M_1} \otimes \cdots \otimes \mathrm{d}x^{M_p]}, \qquad \text{e.g.} \qquad \mathrm{d}x \wedge \mathrm{d}y = \mathrm{d}x \otimes \mathrm{d}y - \mathrm{d}y \otimes \mathrm{d}x, \tag{4.17}$$

with the antisymmetrisation of the conventions. The products $\mathrm{d}x^{M_1} \wedge \cdots \wedge \mathrm{d}x^{M_p}$ with $M_1 < \cdots < M_p$ form a basis of the p-forms, and the components $C_{M_1 \cdots M_p}$ are the coefficients of the expansion in this basis. In four dimensions, for instance, the two-forms have the basis $\mathrm{d}x^1 \wedge \mathrm{d}x^2$, $\mathrm{d}x^1 \wedge \mathrm{d}x^3$, …, with $\binom{4}{2} = 6$ elements, each of which can be seen as an antisymmetric 4×4 matrix with $+1$ in one entry above the diagonal, -1 in the transposed entry and zeros elsewhere.

4.4.1 Three ways of looking at $\mathrm{d}x \wedge \mathrm{d}y$

Three meanings of an object such as $\mathrm{d}x \wedge \mathrm{d}y$ are useful.

(1) Algebraically, as an antisymmetric tensor. Represent $\mathrm{d}x$ and $\mathrm{d}y$ by the basis vectors $(1,0)^T$ and $(0,1)^T$. Then

$$\mathrm{d}x \wedge \mathrm{d}y = \begin{pmatrix}1\\0\end{pmatrix}\begin{pmatrix}0 & 1\end{pmatrix} - \begin{pmatrix}0\\1\end{pmatrix}\begin{pmatrix}1 & 0\end{pmatrix} = \begin{pmatrix}0 & 1\\-1 & 0\end{pmatrix}. \tag{4.18}$$

As a tensor with two lower indices it acts on two vectors $\vec{a}$ and $\vec{b}$ and gives a number,

$$(\mathrm{d}x \wedge \mathrm{d}y)(\vec{a},\vec{b}) = \begin{pmatrix}a_x & a_y\end{pmatrix}\begin{pmatrix}0 & 1\\-1 & 0\end{pmatrix}\begin{pmatrix}b_x\\b_y\end{pmatrix} = a_x b_y - a_y b_x = \det(\vec{a},\vec{b}), \tag{4.19}$$

the determinant of the matrix whose columns are $\vec{a}$ and $\vec{b}$.

(2) Geometrically, as an oriented infinitesimal parallelogram, the one spanned by $\mathrm{d}x$ and $\mathrm{d}y$ (Figure 4.4(a)). The orientation matters: $\mathrm{d}y \wedge \mathrm{d}x = -\mathrm{d}x \wedge \mathrm{d}y$ is the same parallelogram with the opposite orientation. Two vectors, identified with the one-forms $\vec{a} = a_x\mathrm{d}x + a_y\mathrm{d}y$ and $\vec{b} = b_x\mathrm{d}x + b_y\mathrm{d}y$, span a parallelogram given by

$$\vec{a} \wedge \vec{b} = (a_x b_y - a_y b_x)\,\mathrm{d}x \wedge \mathrm{d}y, \tag{4.20}$$

since $\mathrm{d}x \wedge \mathrm{d}x = \mathrm{d}y \wedge \mathrm{d}y = 0$ by antisymmetry. The coefficient is the oriented area of the parallelogram (Figure 4.4(b)), the same determinant as in Equation (4.19).

(3) As an oriented volume element, which fixes the measure of an integral. This is the meaning that matters for actions such as Equation (4.13), obtained by integrating forms. A two-form in two dimensions, with coordinates $(x^1, x^2) = (x, y)$, is

$$C_2 = \frac{1}{2}\,C_{MN}\,\mathrm{d}x^M \wedge \mathrm{d}x^N = \frac{1}{2}\big(C_{12}\,\mathrm{d}x \wedge \mathrm{d}y + C_{21}\,\mathrm{d}y \wedge \mathrm{d}x\big) = C_{12}\,\mathrm{d}x \wedge \mathrm{d}y,$$

since both terms are equal, and $\mathrm{d}x \wedge \mathrm{d}y$ is read as the measure $\mathrm{d}x\,\mathrm{d}y$, keeping track of the orientation. A two-form is therefore integrated over a two-dimensional surface Σ_2 (a sphere, or any other), and

$$\int_{\Sigma_2} C_2 = \int_{\Sigma_2} C_{12}(x,y)\,\mathrm{d}x\,\mathrm{d}y \tag{4.21}$$

for coordinates (x, y) of positive orientation on Σ_2. In the same way, in terms of the pulled-back components Equation (4.15), the Chern–Simons action Equation (4.13) is T_p times the integral of $C_{01\cdots p}(\xi)$ over $\mathrm{d}^{p+1}\xi$.

4.4.2 Exterior derivative and Hodge star

The *exterior derivative* maps a p-form into a $(p+1)$-form,

$$\mathrm{d}C_p = \frac{1}{p!}\,\partial_{M_0} C_{M_1\cdots M_p}(x)\;\mathrm{d}x^{M_0} \wedge \mathrm{d}x^{M_1} \wedge \cdots \wedge \mathrm{d}x^{M_p}\,: \tag{4.22}$$

the derivative brings one more index, and the result is expanded in the basis of $(p+1)$-forms. Applying it twice gives zero,

$$\mathrm{d}^2 = 0, \tag{4.23}$$

because $\mathrm{d}^2 C_p$ contains $\partial_N \partial_{M_0} C_{M_1\cdots M_p}$, which is symmetric under the exchange of N and M_0, contracted with $\mathrm{d}x^N \wedge \mathrm{d}x^{M_0}$, which is antisymmetric.

The other basic operation is *Hodge duality*, a map $\star$ from p-forms to $(D-p)$-forms on a D-dimensional manifold with metric g_{MN}. On a basis element it is

$$\star\left(\mathrm{d}x^{M_1} \wedge \cdots \wedge \mathrm{d}x^{M_p}\right) = \frac{1}{(D-p)!}\,\sqrt{|g|}\;g^{M_1N_1}\cdots g^{M_pN_p}\;\varepsilon_{N_1\cdots N_pN_{p+1}\cdots N_D}\;\mathrm{d}x^{N_{p+1}} \wedge \cdots \wedge \mathrm{d}x^{N_D}, \tag{4.24}$$

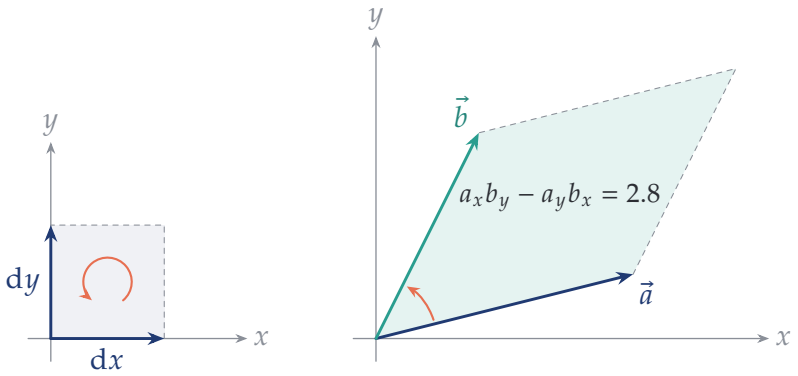


Figure 4.4: The two-form $\mathrm{d}x \wedge \mathrm{d}y$ as an oriented infinitesimal parallelogram. (a) The parallelogram spanned by $\mathrm{d}x$ and $\mathrm{d}y$, with its orientation. (b) Two vectors $\vec{a}$ and $\vec{b}$ span a parallelogram whose oriented area is $a_x b_y - a_y b_x$, Equation (4.20); exchanging them reverses the orientation and the sign.

and it acts on a general p-form through its basis elements. The second indices of the p inverse metrics are contracted with the first p indices of the Levi-Civita symbol ε, and its remaining $D - p$ indices with the new basis elements; the sign of ε and the meaning of $\sqrt{|g|}$ are fixed in the conventions.

With the Hodge star one defines an inner product of two p-forms A_p and B_p, by integrating over X the D-form $A_p \wedge \star B_p$:

$$(A_p, B_p) = \int_X A_p \wedge \star B_p = \frac{1}{p!} \int_X \mathrm{d}^D x \, \sqrt{|g|} \, A_{M_1 \cdots M_p} B^{M_1 \cdots M_p}. \tag{4.25}$$

The second form follows by substituting the definitions and contracting the Levi-Civita symbols. The inner product is symmetric, and positive definite when the metric is Riemannian, as on the compact space Y_6. For a Lorentzian metric the components with a time index contribute with the opposite sign, and the product is indefinite.

4.5 Maxwell's theory in the language of forms

In this language electromagnetism is written very compactly. The potential is the one-form $A_1 = A_\mu \, \mathrm{d}x^\mu$, and its field strength the two-form

$$F_2 = \mathrm{d}A_1 = \frac{1}{2} F_{\mu\nu} \, \mathrm{d}x^\mu \wedge \mathrm{d}x^\nu, \qquad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu.$$

In four dimensions the Hodge dual of a two-form is again a two-form, so $F_2 \wedge \star F_2$ is a four-form, which can be integrated over spacetime. Coupled to a current, written as the one-form $J = J_\mu \, \mathrm{d}x^\mu$, the action is

$$S = -\frac{1}{2} \int F_2 \wedge \star F_2 + \int A_1 \wedge \star J = \int \mathrm{d}^4 x \, \sqrt{-g} \left(-\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + A_\mu J^\mu \right), \tag{4.26}$$

where the second form follows from Equation (4.25): the Maxwell action is $-\frac{1}{2}$ times the norm of F_2. Maxwell's equations with sources are

$$-\partial_\mu F^{\mu\nu} = J^\nu \qquad \Longleftrightarrow \qquad \mathrm{d} \star F_2 = \star J, \tag{4.27}$$

an equality between three-forms: starting from F_2, the star gives a two-form and d a three-form, while the star of the one-form J is a three-form.

The other two Maxwell equations are the *Bianchi identity*,

$$\partial_{[\mu}F_{\rho\sigma]} = 0 \qquad \Longleftrightarrow \qquad \partial_\mu(\star F)^{\mu\nu} = 0 \qquad \Longleftrightarrow \qquad \mathrm{d}F_2 = 0, \tag{4.28}$$

with $(\star F)_{\mu\nu} = \frac{1}{2}\sqrt{-g}\,\varepsilon_{\mu\nu\rho\sigma}F^{\rho\sigma}$ the components of $\star F_2$ (the middle form is specific to four dimensions, where $\star F_2$ is a two-form). In the language of forms it is satisfied automatically: if $F_2 = \mathrm{d}A_1$, then $\mathrm{d}F_2 = \mathrm{d}^2A_1 = 0$ by Equation (4.23). Conversely, $\mathrm{d}F_2 = 0$ implies that, at least locally, F_2 can be written as $\mathrm{d}A_1$, i.e. $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$.

4.6 Electric–magnetic duality and the Dirac quantisation condition

To get to branes one more connection is needed. In the absence of sources Maxwell's equations are

$$\mathrm{d}\star F_2 = 0 = \mathrm{d}F_2, \tag{4.29}$$

and they are invariant under the exchange of F_2 with its Hodge dual, $F_2 \to \star F_2$ (then $\star F_2 \to \star\star F_2 = -F_2$, which is equally closed). In terms of the electric and magnetic fields, this exchanges the two:

$$F_2 \longrightarrow \star F_2 \qquad \Longleftrightarrow \qquad (\vec{E}, \vec{B}) \longrightarrow (-\vec{B}, \vec{E}). \tag{4.30}$$

Since the equations of motion are unchanged, the theory looks the same after the exchange: this is *electric–magnetic duality*.

With sources the symmetry is lost: the current $\star J$ appears in one equation and not in the other, because there are electrically charged particles but no magnetically charged ones, or at least none has been seen so far, perhaps because they are too heavy to have been produced. The duality survives if one introduces magnetic charges, *magnetic monopoles*, besides the electric ones. The reason for insisting on this is that branes, as we shall see, always come with both electric and magnetic charges. Let

$$J^\mu = (\rho_e, \vec{J}), \qquad K^\mu = (\rho_m, \vec{K})$$

be the electric and magnetic four-currents, with charge densities ρ_e, ρ_m and three-currents $\vec{J}$, $\vec{K}$. Maxwell's equations become

$$-\partial_\mu F^{\mu\nu} = J^\nu, \quad \partial_\mu(\star F)^{\mu\nu} = K^\nu \qquad \Longleftrightarrow \qquad \mathrm{d}\star F_2 = \star J, \quad \mathrm{d}F_2 = \star K, \tag{4.31}$$

with $K = K_\mu\,\mathrm{d}x^\mu$, so that $\nabla\cdot\vec{B} = \rho_m$, and now the theory looks symmetric. Indeed, the transformation

$$\big(F_2,\ \star F_2\big) \longrightarrow \big(\star F_2,\ -F_2\big), \qquad \big(J,\ K\big) \longrightarrow \big(-K,\ J\big) \tag{4.32}$$

maps the two equations into each other: the first becomes $\mathrm{d}(-F_2) = -\star K$, which is the second, and the second becomes $\mathrm{d}\star F_2 = \star J$, which is the first. Exchanging the fields and the two kinds of charged particles maps the theory into one that looks exactly the same: it is the same theory, seen from a different point of view.

4.6.1 The Dirac quantisation condition

Why are electric charges quantised? All the charges seen in nature are integer multiples of a basic unit: if the electron has charge -1 in units of e, the quarks have charges $\frac{2}{3}$ and $-\frac{1}{3}$, fractional but still multiples of $\frac{1}{3}$. We do not know the reason, but magnetic monopoles, if they exist, would explain it. The argument is quantum mechanical. Consider a particle of electric charge q_e moving around a monopole of magnetic charge q_m. Going around a closed loop its wave function picks up a phase, which must be trivial for the wave function to be single-valued. This requires the *Dirac quantisation condition*

$$q_e\, q_m = 2\pi n, \qquad n \in \mathbb{Z} \tag{4.33}$$

(in the units of the conventions, $\hbar = 1$ and Heaviside–Lorentz charges): the product of the electric and the magnetic charge is an integer multiple of 2π [14]. It explains charge quantisation: in the presence of a monopole of charge q_m, every electric charge must be an integer multiple of $2\pi/q_m$. That charges are quantised in nature is therefore a strong hint that monopoles exist.

The same condition suggests why monopoles are so hard to detect. At fixed n the magnetic charge is inversely proportional to the electric one, $q_m = 2\pi n/q_e$, so if the electric charge is small the magnetic charge is large: if one kind of object is weakly coupled, the other is strongly coupled. Electric–magnetic duality, Equation (4.32), is therefore a duality between weak and strong coupling, the prototype of the S-duality of Section 2.2. The next lecture generalises these ideas to branes.

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